12/4/2026

Business Report

Capstone Project – Final Submission

Sonal Kamble

Table of Contents

Problem Statement – Agent Bonus prediction 4

Business Context: 4

Objective: 4

Data Description: 4

Problem Understanding: 5

Need of the study/project 5

Understanding business/social opportunity 5

Overview of the dataset: 6

Shape of the dataset- 6

Data types of the columns for the dataset 6

Statistical summary of the dataset 6

Duplicate values: 6

Missing Values: 6

Data Cleaning 6

Exploratory Data Analysis (EDA) 7

A. Univariate Analysis of Agent Bonus: 7

B. Univariate Analysis of Age: 8

C. Univariate Analysis of CustTenure: 9

D. Univariate Analysis of ExistingProdType: 10

E. Univariate Analysis of NumberOfPolicy: 11

F. Univariate Analysis of MonthlyIncome 12

G. Univariate Analysis of Complaint 13

H. Univariate Analysis of Existing Policy Tenure 14

I. Univariate Analysis of LastMonthCalls 15

J. Univariate Analysis of Cust Care Score 16

- Univariate Analysis of Categorical Variables 17

- Multivariate Analysis 24

- Bivariate Analysis of the target variable vs the numerical variable 25

I. Age vs AgentBonus 25

II. CustTenure vs AgentBonus 26

III. SumAssured vs AgentBonus 27

IV. Monthly Income Vs AgentBonus 28

➢ Bivariate analysis of the Target Variable vs categorical variables 29
a) Channel vs. Agent Bonus 29
b) Occupation vs. Agent Bonus 30
c) Designation Vs AgentBonus 31
Data Preprocessing 31
Duplicate value check and treatment 31
Anomalous value check and treatment 31
Missing value check and treatment 31
Outlier check and treatment 33
Feature Engineering 34
Data preparation for modeling 34
Model Building - Baseline Model 35
Performance of the model on multiple metrics 36
Linear Regression Model Training performance 36
Linear Regression Model Training performance 36
Performance Observation: 36
Model Building - Advanced Models 36
Decision Tree Model Regressor: 36
Bagging Regressor: 37
Random Forest Model Regressor: 37
AdaBoost Regressor: 38
Gradient Boosting Regressor 38
XGBoost Regressor: 39
Model Performance Improvement using Hyperparameter Tuning 39
Hyperparameter Tuning - Decision Tree: 40
Hyperparameter Tuning - Random Forest: 40
Hyperparameter Tuning - Gradient Boosting Regressor: 41
Model Performance Comparison and Final Model Selection 41
Checking important features of the Random Forest Tuned Model 42
Observed VS Predicted Values Plot of the test data for the Random Forest Model (Tuned) 43
Key insights from the analysis conducted 44
Actionable insights and business recommendations 45
Conclusion 45

List Of Tables

Table 1: Statistical Summary 6
Table 2:Boxplot and histogram of Agent Bonus 7
Table 3:Boxplot and histogram of Age 8
Table 4:Boxplot and Histogram of CustTenure 9
Table 5:Boxplot and Histogram of ExistingProdType 10
Table 6:Boxplot and Histogram of NumberOfPolicy 11
Table 7:Boxplot and Histogram of MonthlyIncome 12
Table 8:Boxplot and Histogram of ExistingPolicy Tenure 14
Table 9:Boxplot and Histogram of LastMonthcalls 15
Table 10:Boxplot and Histogram of CustcareScore 16
Table 11:Labeled Barplot of Channel 17
Table 12:Labeled Barplot of Occupation 18
Table 13:Labeled Barplot of EducationField 19
Table 14:Labeled Barplot of Gender 20
Table 15:Labeled Barplot of Designation 21
Table 16: Labeled Barplot of Marital Status 22
Table 17:Labeled Barplot of Zone 22
Table 18:Labeled Barplot of PaymentMethod 23
Table 19:Correlation Analysis Heatmap 24
Table 20:Scatterplot of Age Vs AgentBonus 25
Table 21: Scatterplot of CustTenure Vs AgentBonus 26
Table 22: Scatterplot of SumAssured Vs AgentBonus 27
Table 23: Scatterplot of Monthly Income Vs AgentBonus 28
Table 24: Scatterplot of Channel Vs AgentBonus 29
Table 25:Boxplot of Occupation Vs Agent Bonus 30
Table 26:Boxplot of Designation Vs AgentBonus 31
Table 27: Missing Values After Treatment 32
Table 28: Missing Values Before Treatment 32
Table 29: Boxplots of all the Numeric Variables 33
Table 30: X and Y Variables 34
Table 31: OLS Baseline Regression Model 35
Table 32: Linear Regression Training Performance 36
Table 33: Linear Regression Testing Performance 36
Table 34: Decision Tree Model performance on train set 37
Table 35: Decision Tree Model Performance on test set 37
Table 36: Bagging Model performance on Training set 37
Table 37:Bagging model performance on test set 37
Table 38:Random Forest Model train performance 37
Table 39: random Forest Model Test performance 38
Table 40: AdaBoost Model Performence on Train set 38
Table 41: AdaBoost Model performance on Test set 38

Table 42: Gradiant Boosting Model performance on train set 38
Table 43: Gradiant Boosting model performance on test set 39
Table 44: XGBoost Model Performance on train set 39
Table 45: XGBoost Model Performance on test set 39
Table 46: Decision Tree Tuned Model performance on Train set 40
Table 47:Decision Tree Tuned Model performance on Test set 40
Table 48: Random Forest Model Tuned performance on Train set 40
Table 49: Random Forest Model tuned model performance on test set 40
Table 50: Gradient Boosting Tuned Model performance on Train set 41
Table 51:Gradient Boosting Tuned Model performance on Test Set 41

Problem Statement – Agent Bonus prediction

Business Context:

A leading life-insurance company pays bonus amounts to its field agents based on their performance in terms of sales and customer interaction attributes. The company wants to better understand and manage the bonuses paid to its field agents. Understanding which agents will receive higher bonuses, and why, helps the business move from reactive, month-by-month reward decisions to a proactive approach that aligns incentives with measurable drivers of sales and customer behavior. This context matters because more targeted engagement and training for agents can improve sales effectiveness, reduce avoidable churn, and ensure the bonus budget is invested where it delivers the greatest business impact.

Objective:

As a data scientist at the life insurance company, you are tasked with building a model to predict agent bonuses based on customer and sales-related factors. This will help identify high-performing agents for tailored engagement programs and highlight those needing upskilling, ensuring that incentives and training efforts directly contribute to improved sales performance and efficient resource allocation.

Data Description:

The available data links agent bonus amounts to customer and interaction-level attributes. The detailed data dictionary is given below:

- **CustID:** Unique customer ID
- **AgentBonus:** Bonus amount given to each agent in the last month
- **Age:** Age of the customer
- **CustTenure:** Tenure of the customer in the organization
- **Channel:** Channel through which the acquisition of the customer is done
- **Occupation:** Occupation of the customer
- **EducationField:** Field of education of the customer
- **Gender:** Gender of the customer
- **ExistingProdType:** Existing product type of the customer

- **Designation:** Designation of the customer in their organization
- **NumberOfPolicy:** Total number of existing policies of a customer
- **MaritalStatus:** Marital status of the customer
- **MonthlyIncome:** Gross monthly income of the customer
- **Complaint:** Indicator of complaints registered in the last month by the customer
- **ExistingPolicyTenure:** Max tenure in all existing policies of the customer
- **SumAssured:** Max of sum assured in all existing policies of the customer
- **Zone:** Customer belongs to which zone in India? Like East, West, North, and South
- **PaymentMethod:** Frequency of payment selected by customer, like Monthly, quarterly, half-yearly, and yearly
- **LastMonthCalls:** Total calls attempted by the company to a customer for cross-sell
- **CustCareScore:** Customer satisfaction score given by the customer in the previous service call

Problem Understanding:

The business problem is to predict the bonus amount given to field agents based on their sales performance and customer-related attributes. The goal is to identify patterns that explain why some agents receive higher bonuses than others.

Need of the study/project

In the life insurance industry, field agents play a crucial role in driving sales and maintaining customer relationships. This study can help the company to analyze how different factors influence agent bonuses and to ensure that rewards are distributed in a fair, consistent, and performance-oriented manner.

- By building a predictive model, the company can:
- Accurately forecast agent bonuses
- Recognize and retain high-performing agents
- Identify agents requiring training and performance improvement
- Improve sales effectiveness and customer engagement
- Support management in making strategic, data-driven decisions

Understanding business/social opportunity

Business opportunity in this project can give a business a key factors which are driving agent bonuses so the company can reward the right behaviors, improve customer targeting, and increase sales efficiency. It also helps reduce unnecessary effort and enables data-driven decision-making. From a social perspective, it ensures a fair and transparent bonus system for agents.

Overview of the dataset:

Shape of the dataset-

- There are 4520 rows and 20 columns in the dataset.

Data types of the columns for the dataset

- Dataset has 12 numeric and 8 data type columns in the data.

Statistical summary of the dataset

	CustID	AgentBonus	Age	CustTenure	ExistingProdType	NumberOfPolicy	MonthlyIncome	Complaint	ExistingPolicyTenure	SumAssured	LastMonthCalls	CustCareScore
count	4.520000e+03	4520.000000	4251.000000	4294.000000	4520.000000	4475.000000	4284.000000	4520.000000	4336.000000	4.366000e+03	4520.000000	4468.000000
mean	7.002260e+06	4077.838274	14.494707	14.469027	3.688938	3.565363	22890.309991	0.287168	4.130074	6.199997e+05	4.626991	3.067592
std	1.304956e+03	1403.321711	9.037629	8.963671	1.015769	1.455926	4885.600757	0.452491	3.346386	2.462348e+05	3.620132	1.382968
min	7.000000e+06	1605.000000	2.000000	2.000000	1.000000	1.000000	16009.000000	0.000000	1.000000	1.685360e+05	0.000000	1.000000
25%	7.001130e+06	3027.750000	7.000000	7.000000	3.000000	2.000000	19683.500000	0.000000	2.000000	4.394432e+05	2.000000	2.000000
50%	7.002260e+06	3911.500000	13.000000	13.000000	4.000000	4.000000	21606.000000	0.000000	3.000000	5.789765e+05	3.000000	3.000000
75%	7.003389e+06	4867.250000	20.000000	20.000000	4.000000	5.000000	24725.000000	1.000000	6.000000	7.582360e+05	8.000000	4.000000
max	7.004519e+06	9608.000000	58.000000	57.000000	6.000000	6.000000	38456.000000	1.000000	25.000000	1.838496e+06	18.000000	5.000000

Table 1: Statistical Summary

- The average agent bonus is 4077, while the minimum is 1605 and maximum is 9608.
- Age has an average of 14.49.
- Cust Tenure averages 14.46, ranging from 2 years to 57 years.
- Monthly Income has an average of 22,890, a minimum 16,009, and a maximum of 38,456.
- NumberOfPolicy averages 3.56, with values between 1 and 6.
- SumAssured has an average of 6.2 lakh, a minimum 1.68 lakh, and a maximum of 18.38 lakh.

Duplicate values:

- There are no duplicate values present in the data.

Missing Values:

- The dataset has missing values.

Data Cleaning

- Removed the CustID column as it does not contribute to the analysis.
- Identified and corrected inconsistencies in the Gender column (e.g., “Fe Male” standardized to “Female”).

- Standardized values in the Occupation column (e.g., corrected variations like “Laarge Business” to “Large Business”).
- Standardized values in the Occupation column (e.g., corrected variations like “Laarge Business” to “Large Business”).

Exploratory Data Analysis (EDA)

A. Univariate Analysis of Agent Bonus:

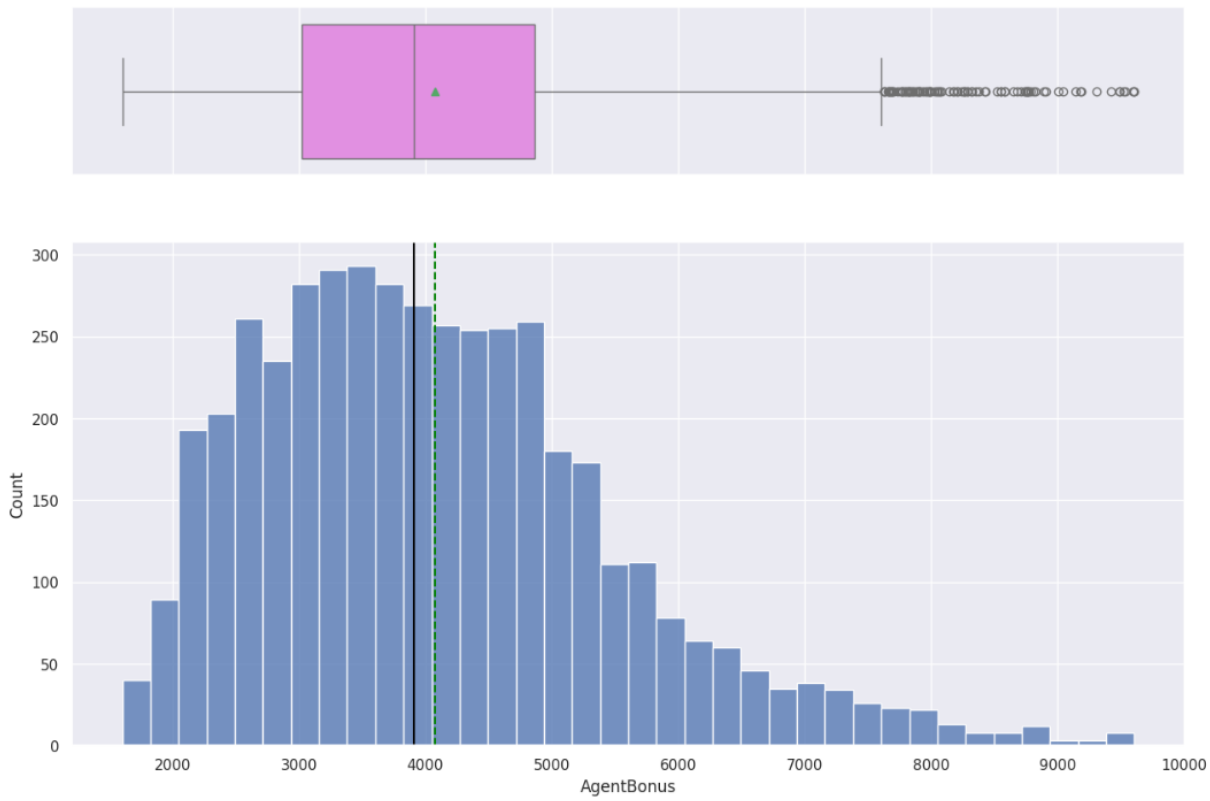

Table2:Boxplot and histogram of Agent Bonus

- The distribution is right skewed and most of the values lies between 2,500 to 5,000 has a long tail towards a higher value upto 9,500.
- There are many outliers are present but they seem to be realistic because if some of the agents performance is high they can earn high bonus.

B. Univariate Analysis of Age:

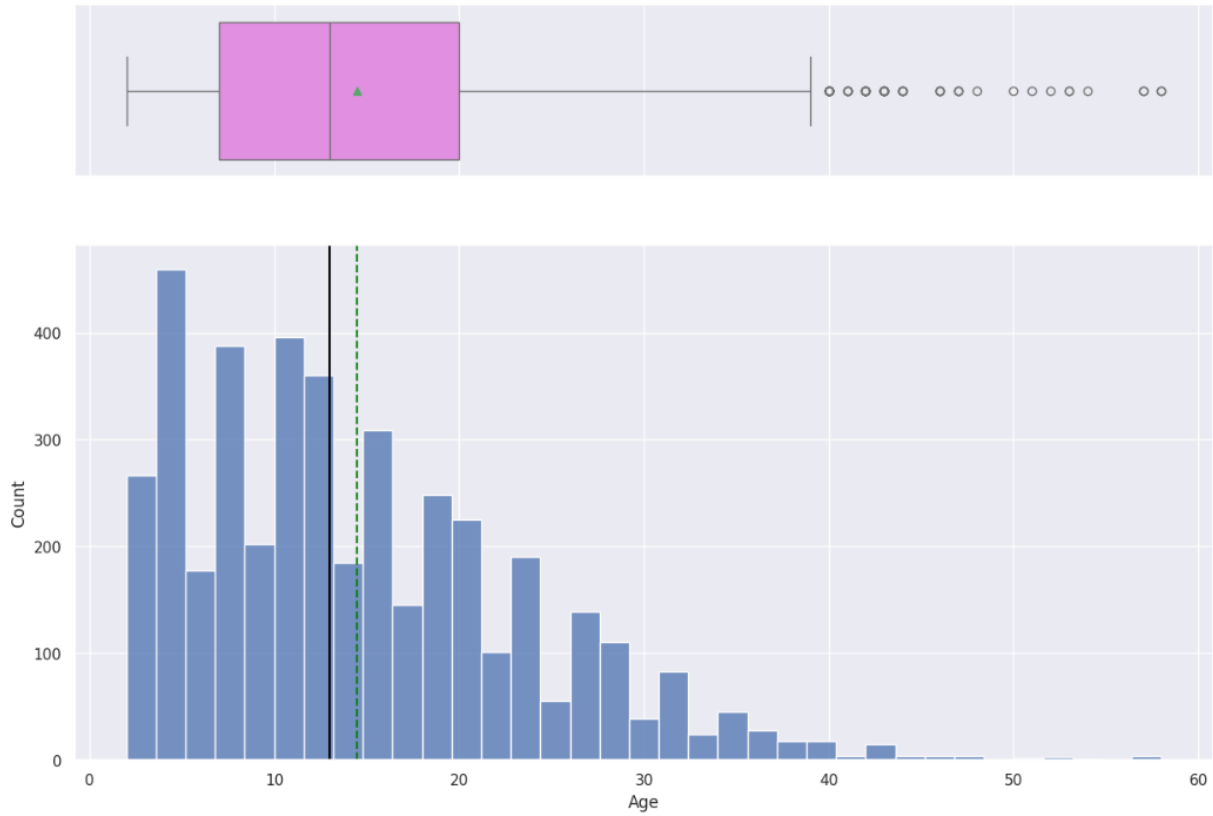

Table3:Boxplot and histogram of Age

- The distribution is right skewed, most of the customer has lower age range between 5 - 20.
- Long tail towards higher age upto 60 and has outliers.
- Very low age (5- 18) seems suspicious and unrealistic, that can also be Policies taken by parents for children so validating the data and not removing outliers because they seem unrealistic.

C. Univariate Analysis of CustTenure:

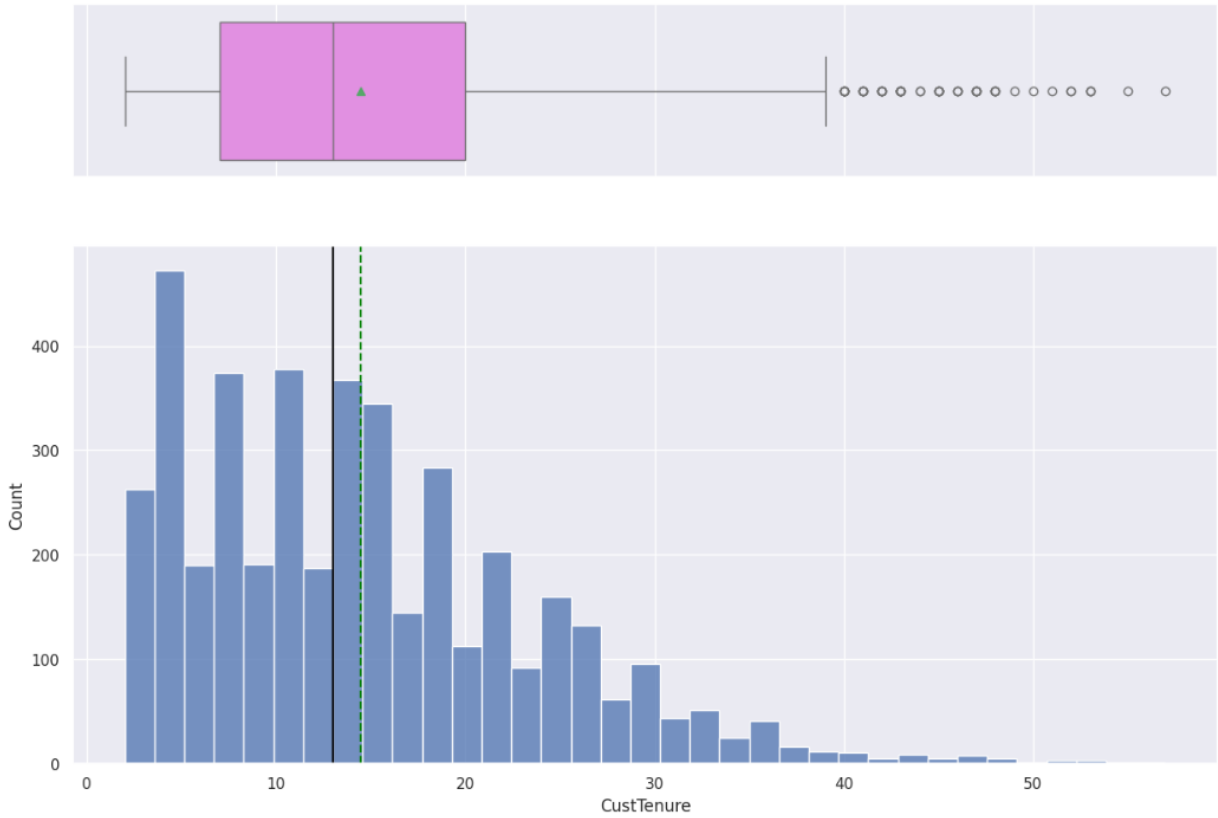

Table4:Boxplot and Histogram of CustTenure

- The distribution is right skewed, most of the customer have tenure between 5 to 20. Observed a long tail exceeding upto 55.
- Some customers have upto 55 years of tenure and it seems to be realistic as the company can be very old or maybe customer have joined very early so not removing outliers.

D. Univariate Analysis of ExistingProdType:

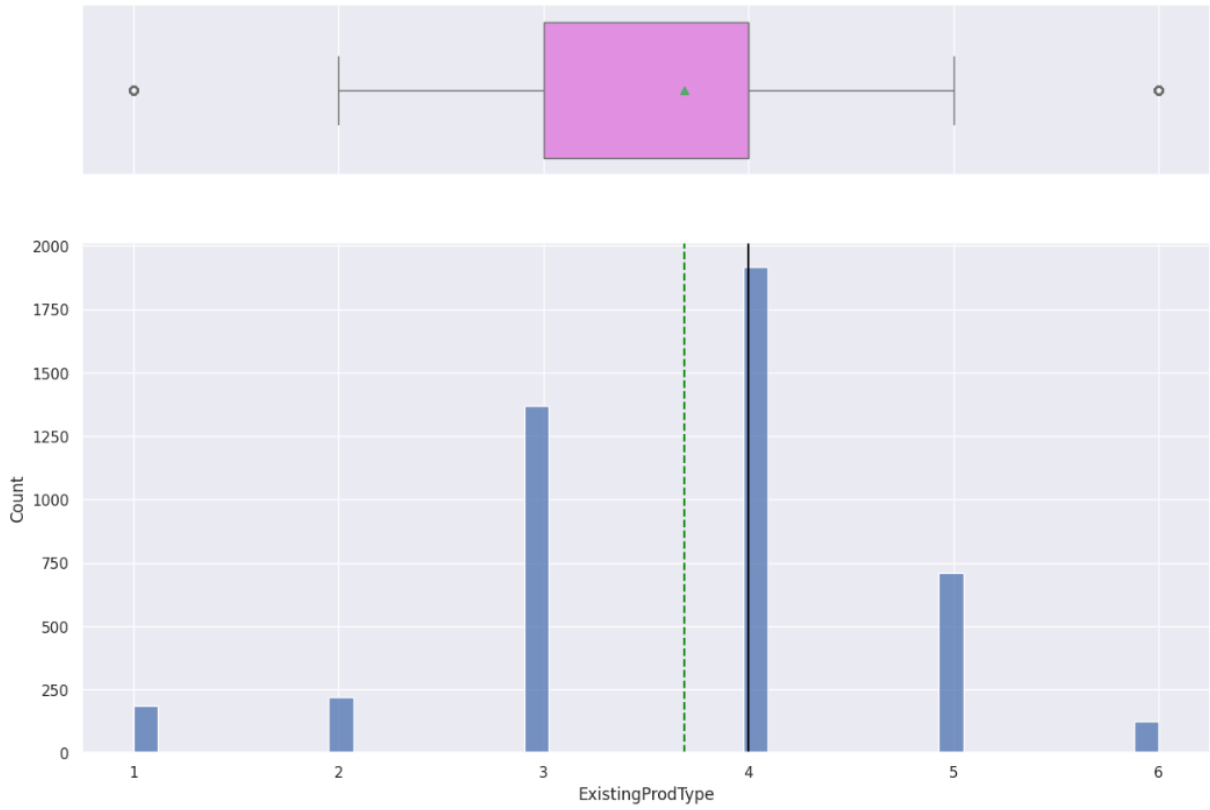

Table 5:Boxplot and Histogram of ExistingProdType

- Product type 4 has the highest count followed by product type 3.

E. Univariate Analysis of NumberOfPolicy:

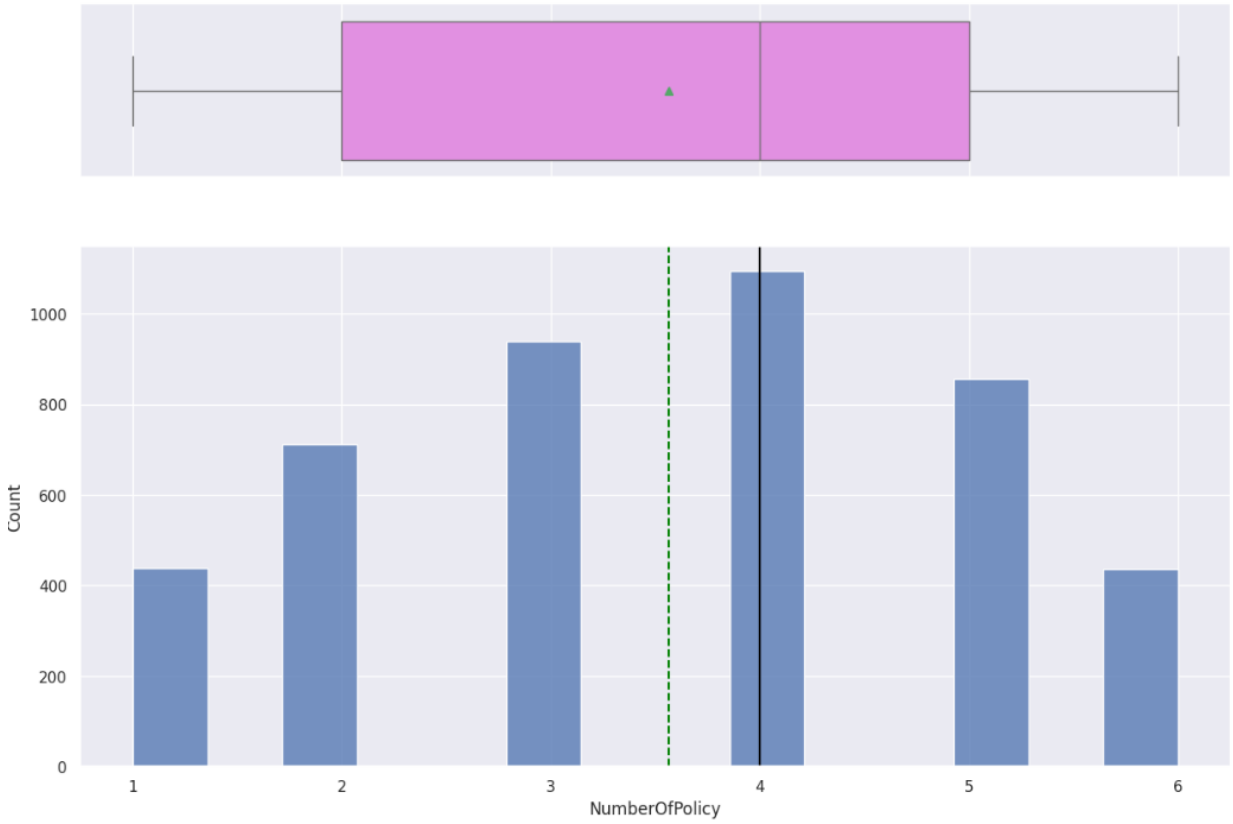

Table 6:Boxplot and Histogram of NumberOfPolicy

- The distribution is slightly left skewed, highest number of policies across the customers.
- No outliers present.

F. Univariate Analysis of MonthlyIncome

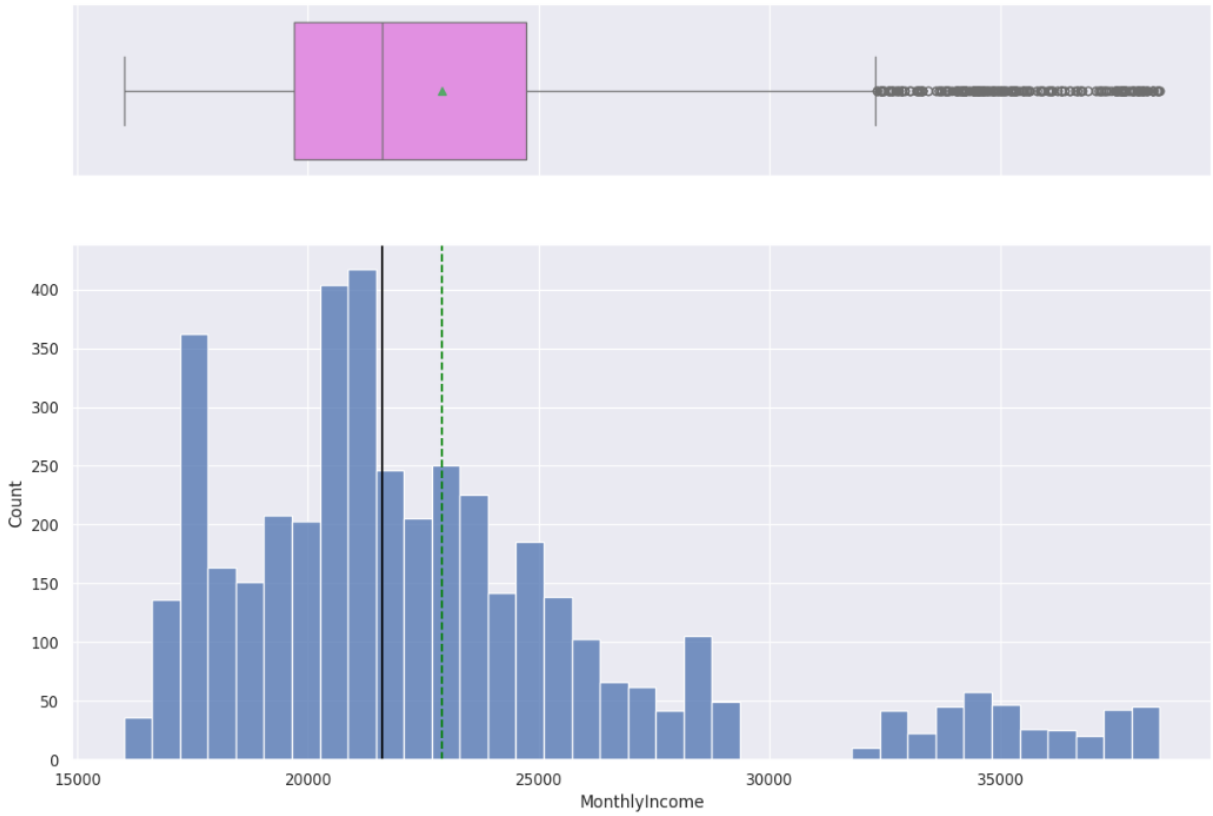

Table 7:Boxplot and Histogram of MonthlyIncome

- The distribution is rightly skewed with a long tail stretching toward the right.
- High income range falls within 20000 to 25000
- High-income outliers ranging from approximately 32,000 to 38,000.
- Keeping those outliers as they are realistic outliers.

G. Univariate Analysis of Complaint

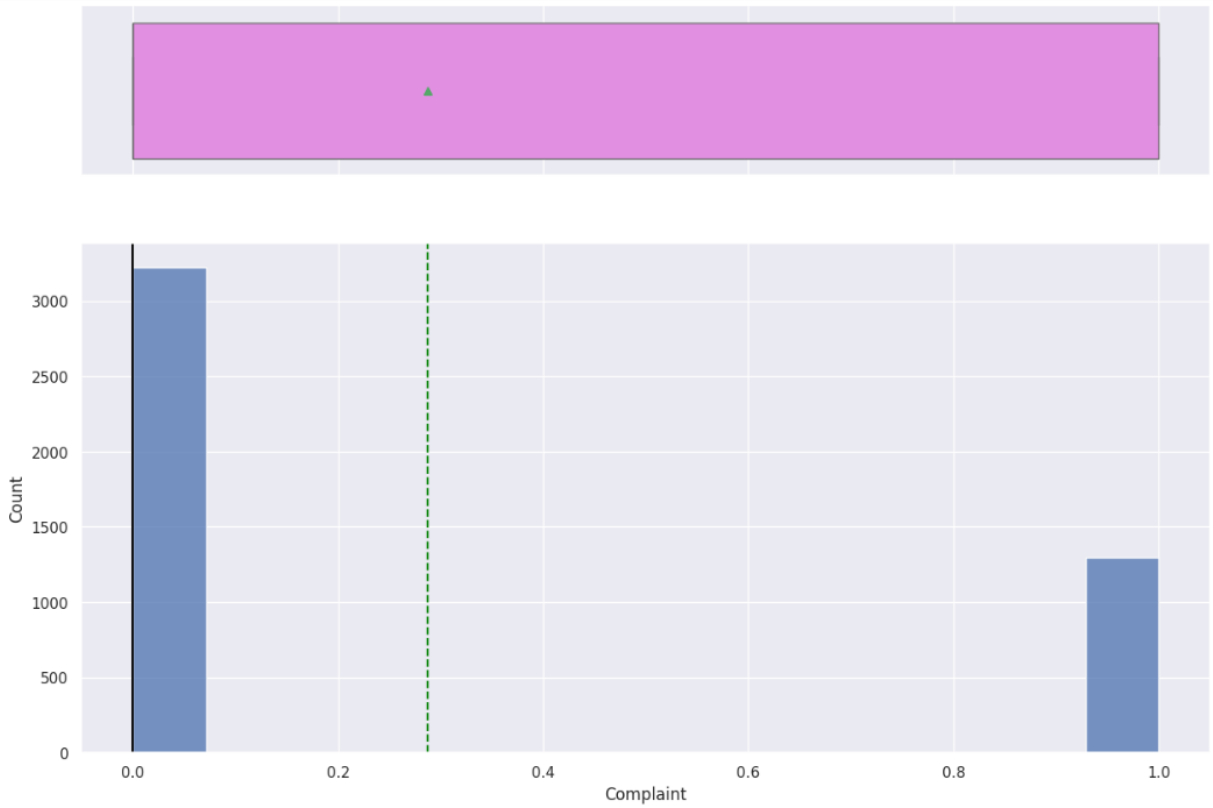

Table 8: Boxplot and Histogram of Complaint

- The majority of the customers did not make any complaints.

H. Univariate Analysis of Existing Policy Tenure

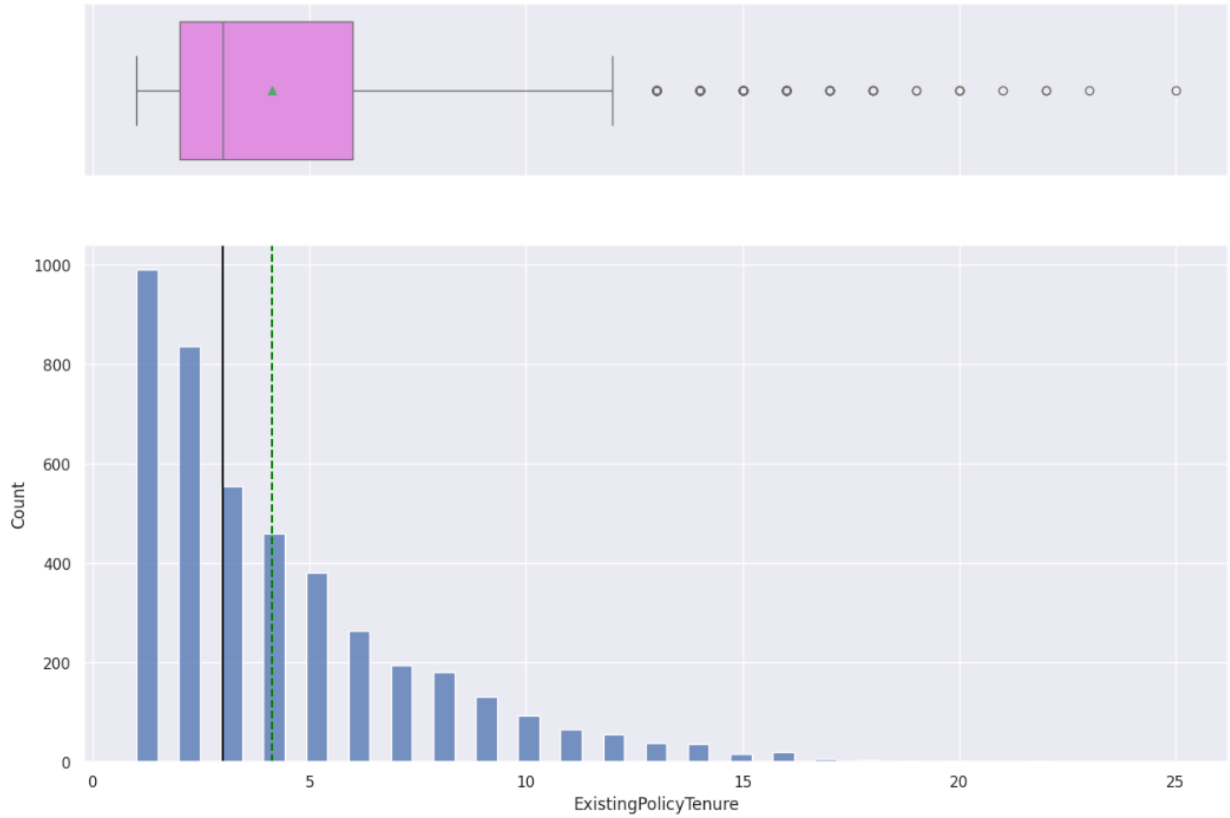

Table 8:Boxplot and Histogram of ExistingPolicy Tenure

- The distribution is highly right skewed and has outliers.
- Most of the customers have been with the policy 2.

I. Univariate Analysis of LastMonthCalls

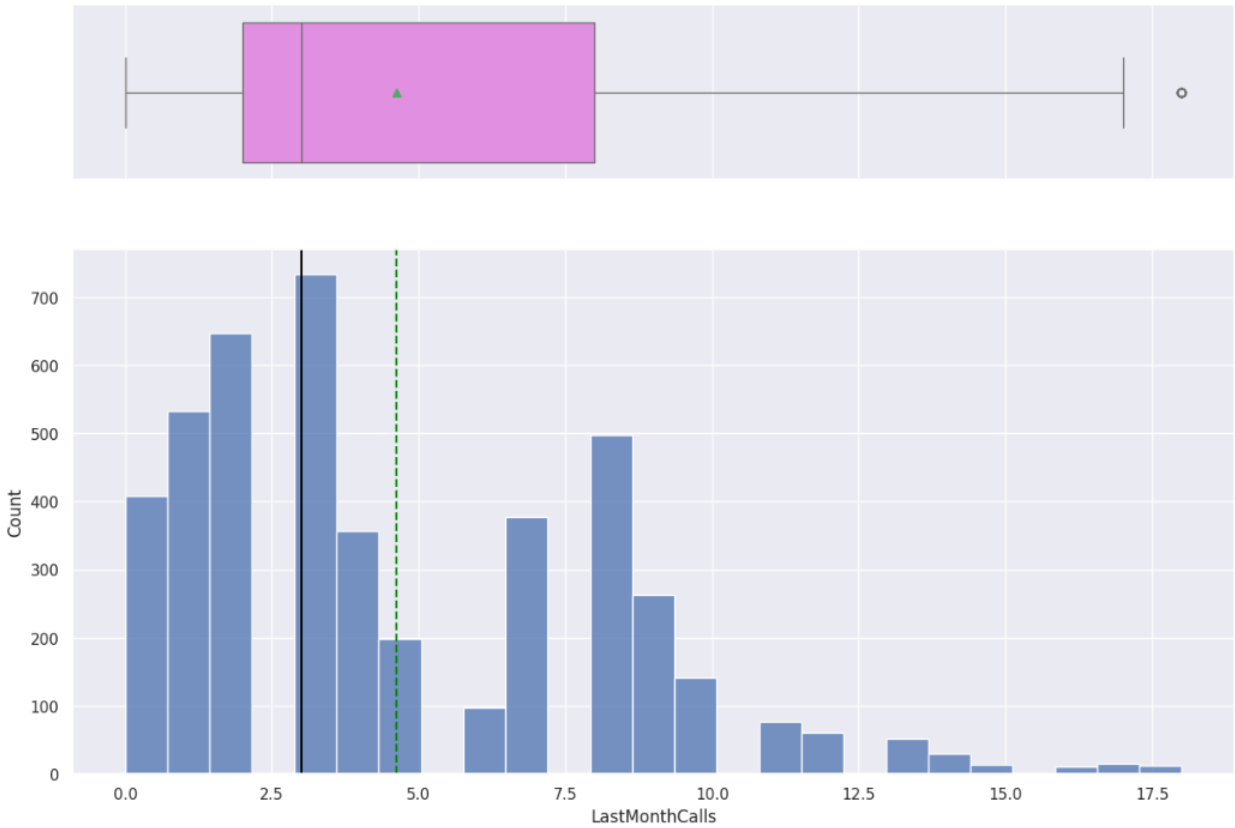

Table 9:Boxplot and Histogram of LastMonthcalls

- The distribution is slightly right skewed, showing a high peak around 3 calls and secondary surge near 8 calls.
- The boxplot confirms one outliers reaching 18 calls.

J. Univariate Analysis of Cust Care Score

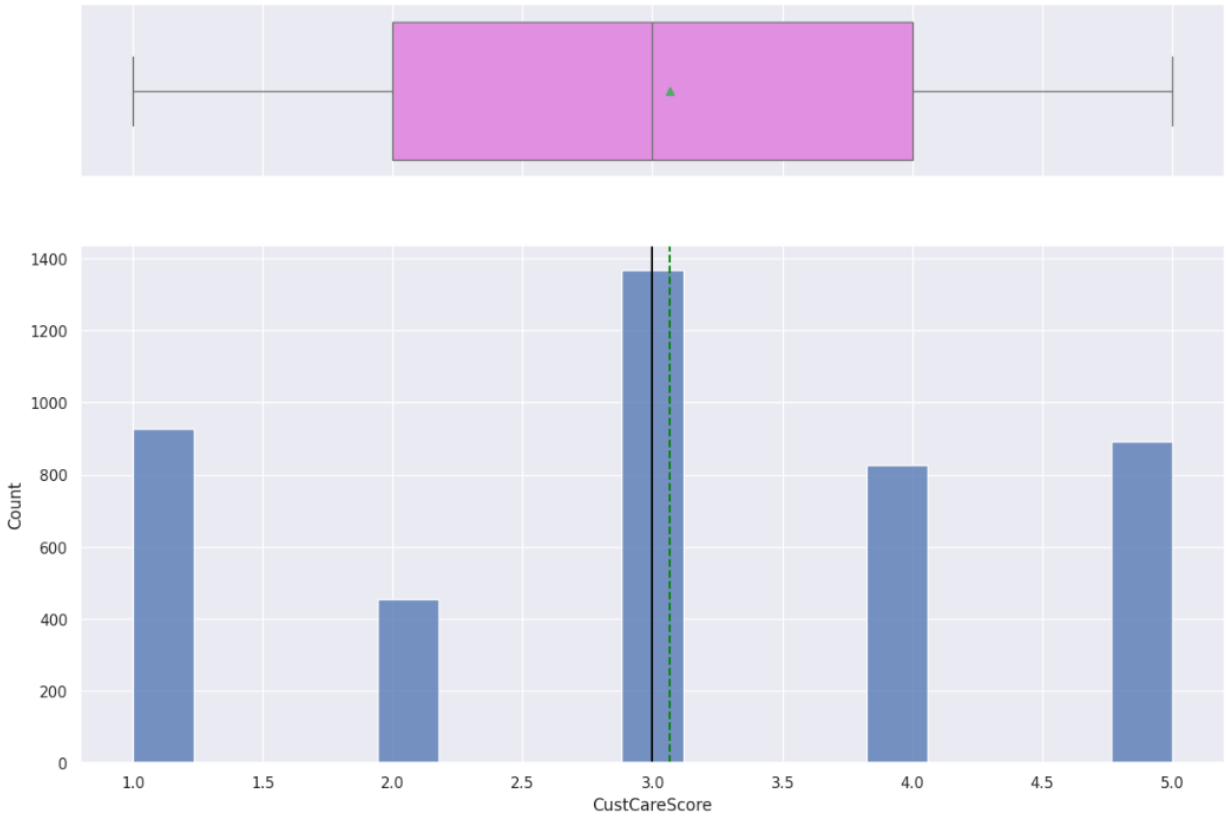

Table 10:Boxplot and Histogram of CustcareScore

- The plot shows discrete distribution with the highest concentration score on 3.
- The boxplot shows a symmetric interquartile range (IQR) between scores 2 and 4, with no outliers detected.

➢ Univariate Analysis of Categorical Variables

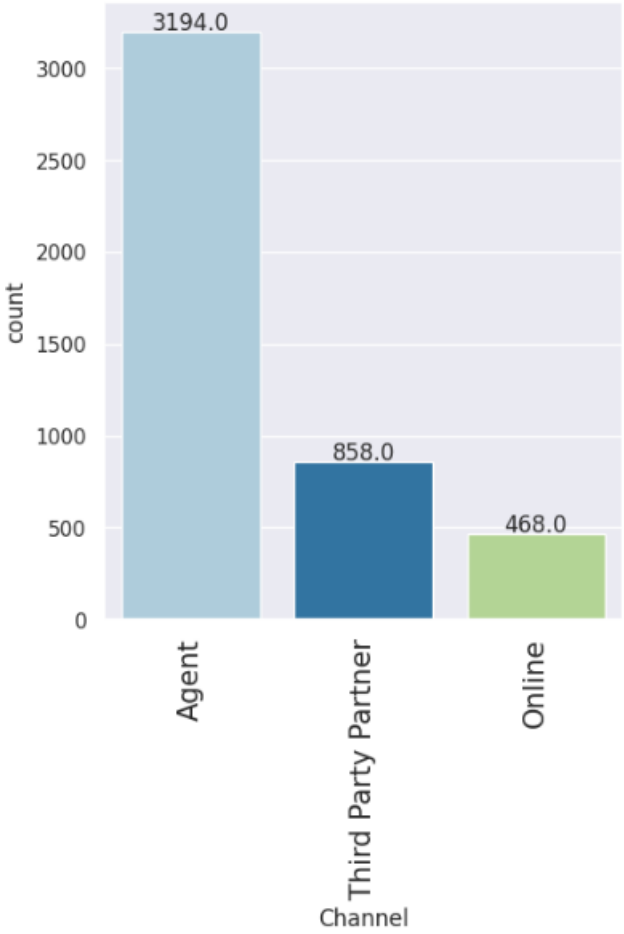

Table 11:Labeled Barplot of Channel

- The agent channel is a dominant source of interaction, with the count of 3,194.
- For third-party partner and Online channel the value count is lower.

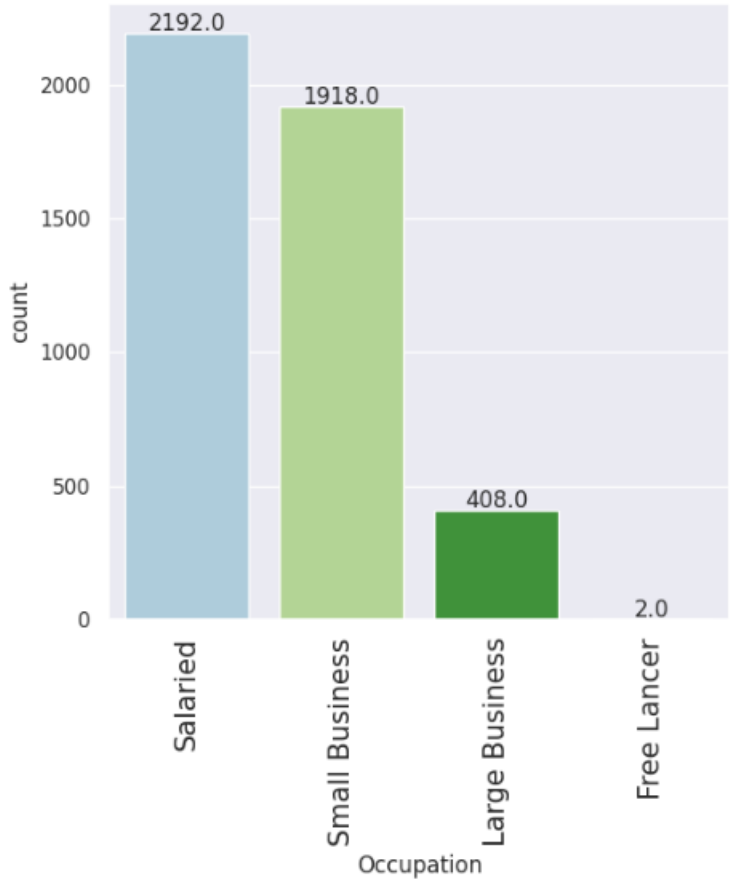

Table 12:Labeled Barplot of Occupation

- In the dataset, the largest portion of customers are salaried (2,192) and have small business (1,918).

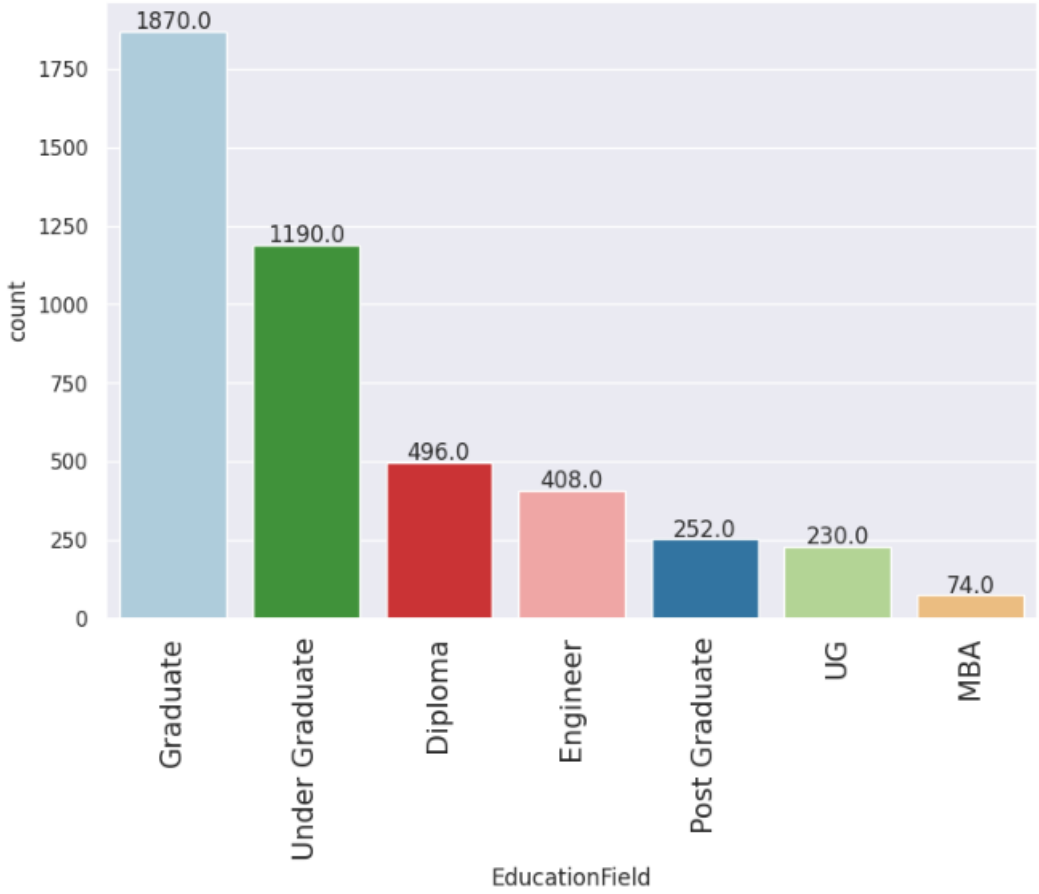

Table 13:Labeled Barplot of EducationField

- The vast majority of the customers consists of Graduates (1,870) and Under Graduates (1,190).

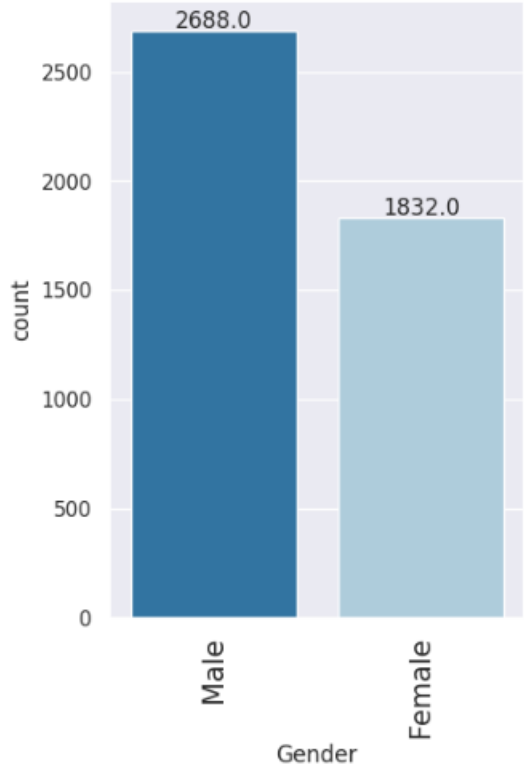

Table 14:Labeled Barplot of Gender

- The majority of the customers who buying policies are Male category followed by female category.

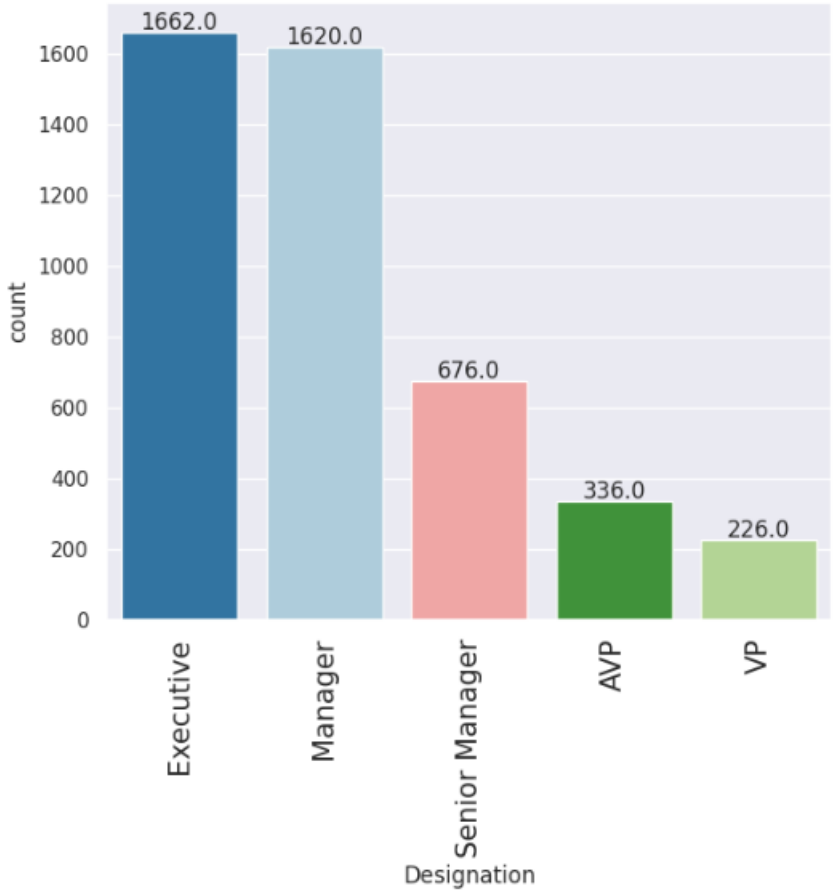

Table 15:Labeled Barplot of Designation

- The largest customer population has designation as Manager(1,620) and Executive (1,535).

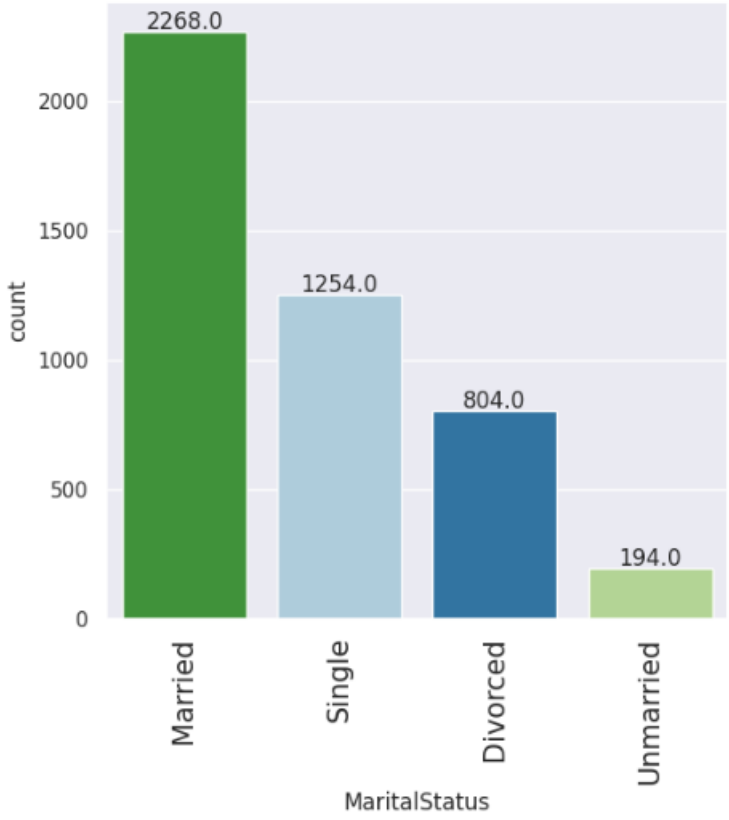

Table 16: Labeled Barplot of Marital Status

- Most of the customers are married with the count of 2,268 followed by single with count 1,254.
- Divorced (804) and Unmarried (194) customers have significantly lower count.

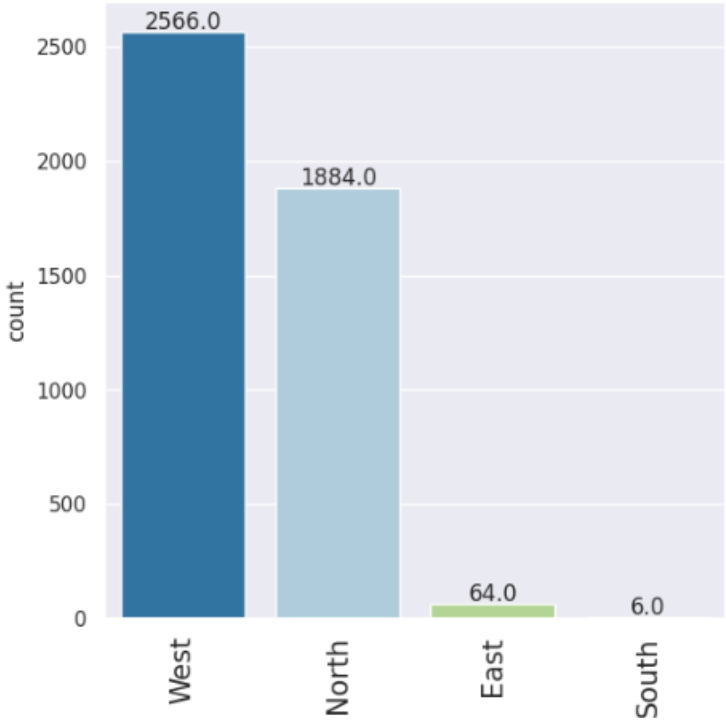

Table 17:Labeled Barplot of Zone

- The dataset is heavily concentrated on West (2,566) and North (1,884) zones.
- For east (64.0) and South (6.0) zones counts are lower.

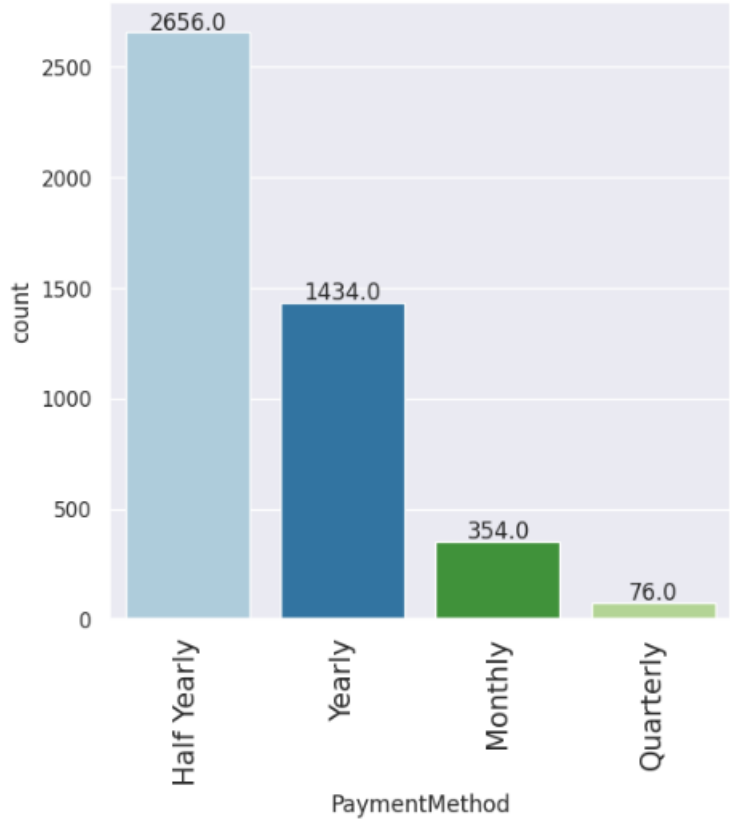

Table 18:Labeled Barplot of PaymentMethod

- Most of the customers are choosing payment method as half yearly with the count of 2,656 followed by Yearly with count of 1,434.

➢ Multivariate Analysis

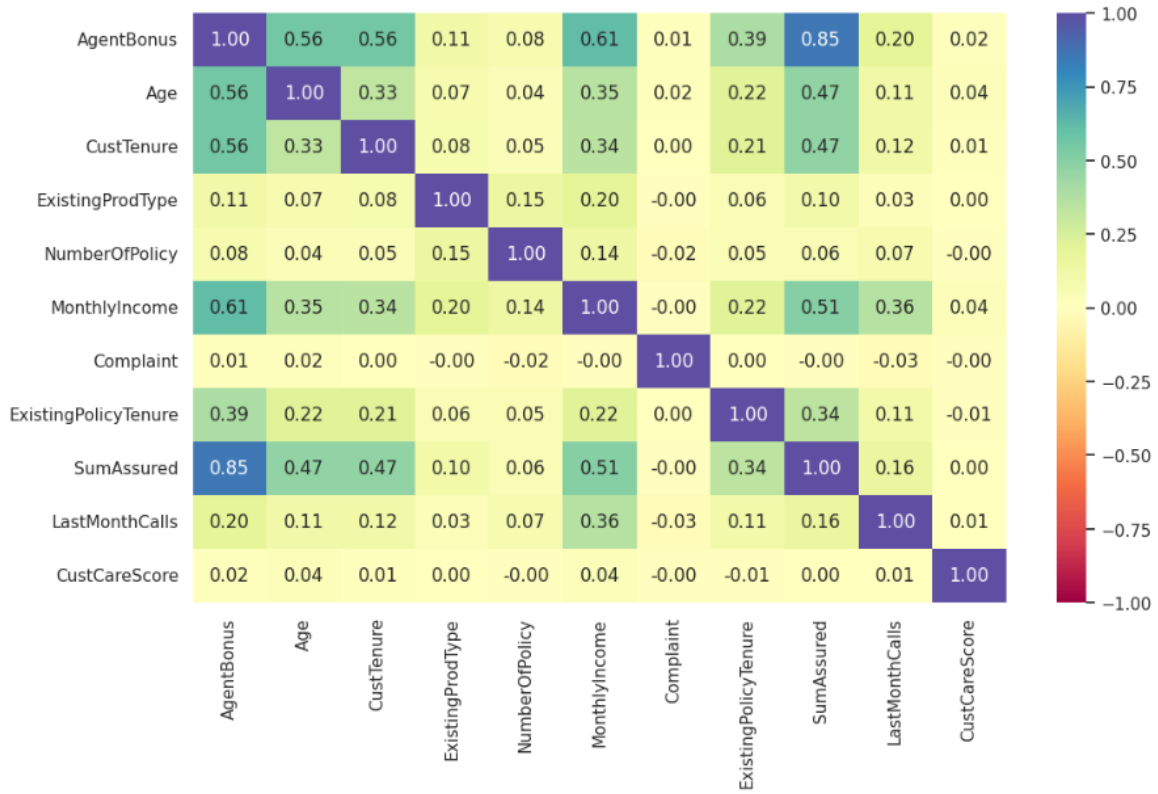

Table 19:Correlation Analysis Heatmap

- AgentBonus has strongly correlated with SumAssured (0.85)
- Secondly agent bonus is also correlated with MonthlyIncome (0.61), Age (0.56), and CustTenure (0.56).
- Monthly income and sum assured is correlated with each other as the customer having high income are particularly choosing policies with higher sum assured.
- Customer care score, number of policy and last month calls has lower correlation.

➢ Bivariate Analysis of the target variable vs the numerical variable

I. Age vs AgentBonus

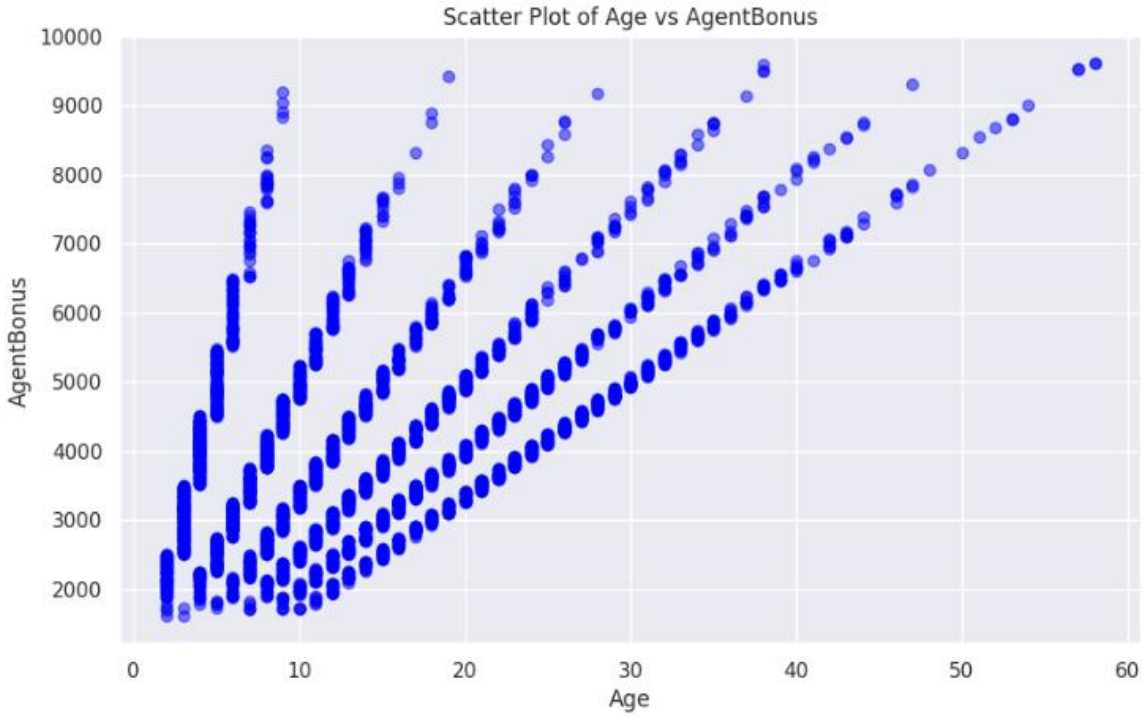

Table 20:Scatterplot of Age Vs AgentBonus

- The plot shows strong positive correlation between Agent Bonus and customer Age as the older customer gets the amount of agent bonus increases.
- The lower age concentrated under age 10, while the highest bonuses (approaching 10,000) are only achievable as Age moves past 40.

II. CustTenure vs AgentBonus

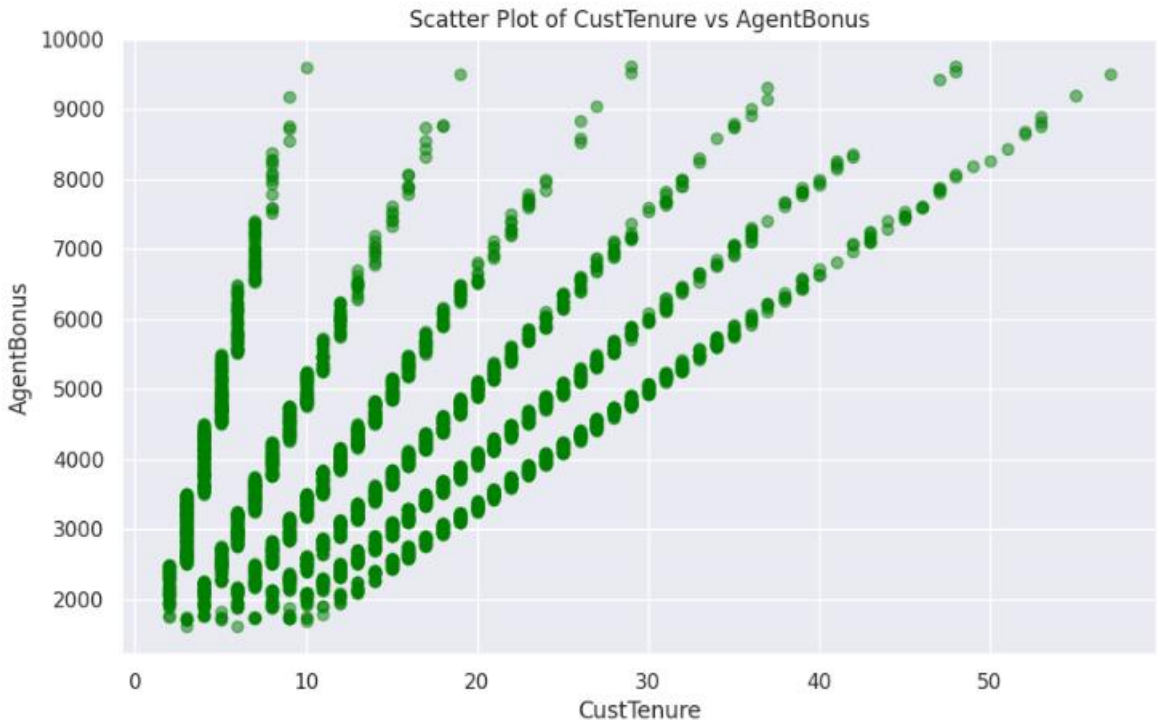

Table 21: Scatterplot of CustTenure Vs AgentBonus

- There is a positive correlation whenever customer tenure increase the agent bonus generally increases.
- The variance increases with tenure. At a tenure of 5, the bonus range is narrow (approx. 2,000 to 4,000), but at a tenure of 30, the range is much wider (approx. 4,000 to 9,500).

III. SumAssured vs AgentBonus

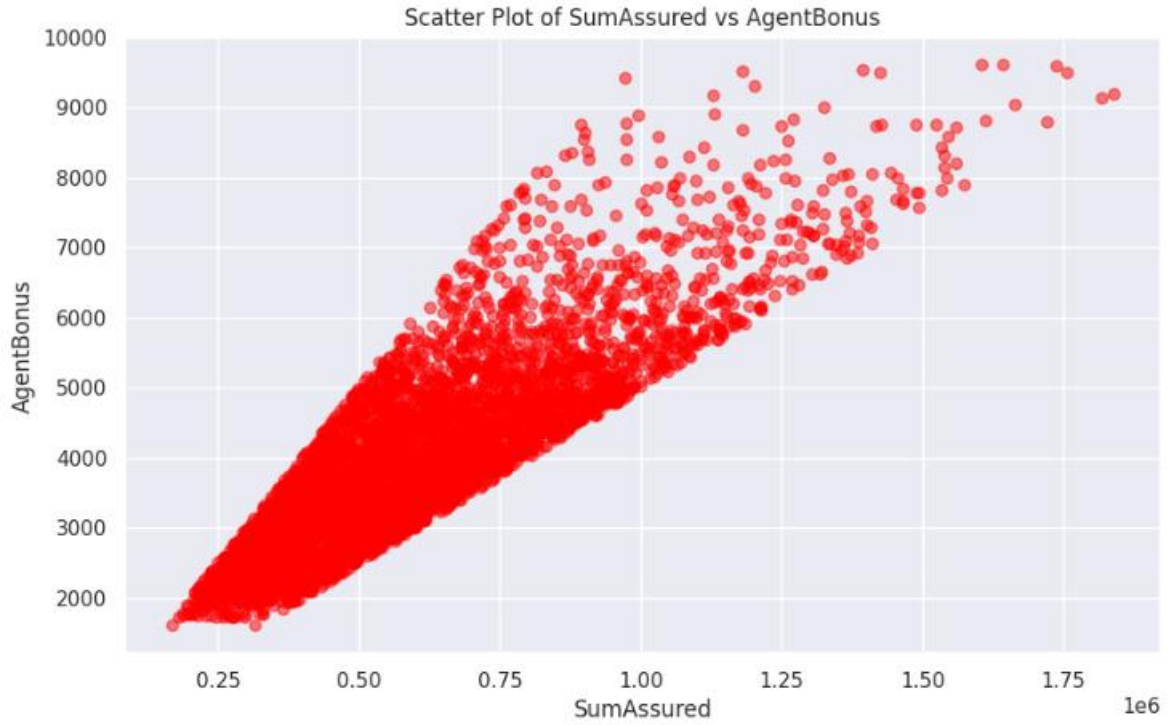

Table 22: Scatterplot of SumAssured Vs AgentBonus

- There is a clear strong positive linear relationship between sum assured and agent bonus as the sum assured increases, the agent bonus tends to increase as well.
- At the lower values of sum assured under 0.50 the data points are tightly clustered.
- As the Sum assured increases, the vertical spread (variance) of AgentBonus expands significantly. This indicates that while the general trend is linear, other factors likely influence the bonus more heavily at higher insurance values.
- The highest concentration of data is in the lower-left quadrant (Sum assured < 0.75M and Agent Bonus < 5000), suggesting the majority of policies fall into this range.
- There are few points at the very high (1.75).

IV. Monthly Income Vs AgentBonus

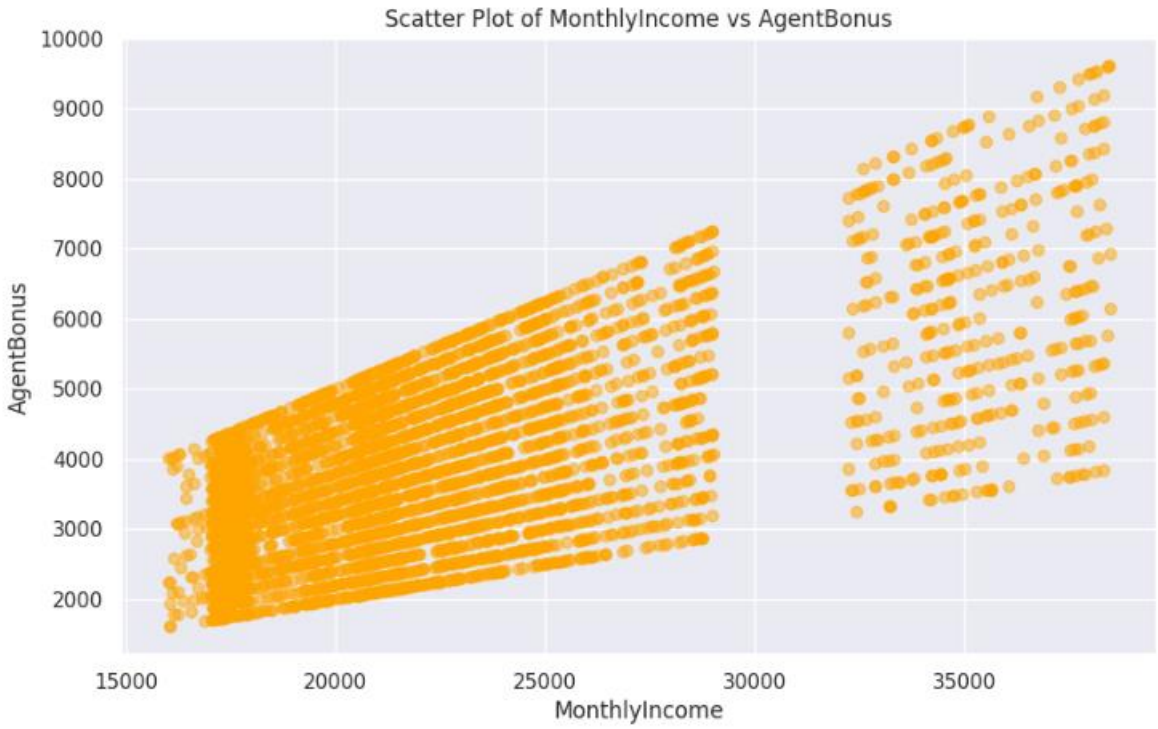

Table 23: Scatterplot of Monthly Income Vs AgentBonus

- There is a clear positive relationship as the monthly income increases agent bonus increases.
- Points form an upward pattern.
- The most data is grouped between income 18,000 to 25,000.
- For higher income values above 30,000 the agent bonus is significantly higher.

➢ Bivariate analysis of the Target Variable vs categorical variables

a) Channel vs. Agent Bonus

```
sns.boxplot(x='Channel', y='AgentBonus', data=data, palette='Set2')
```

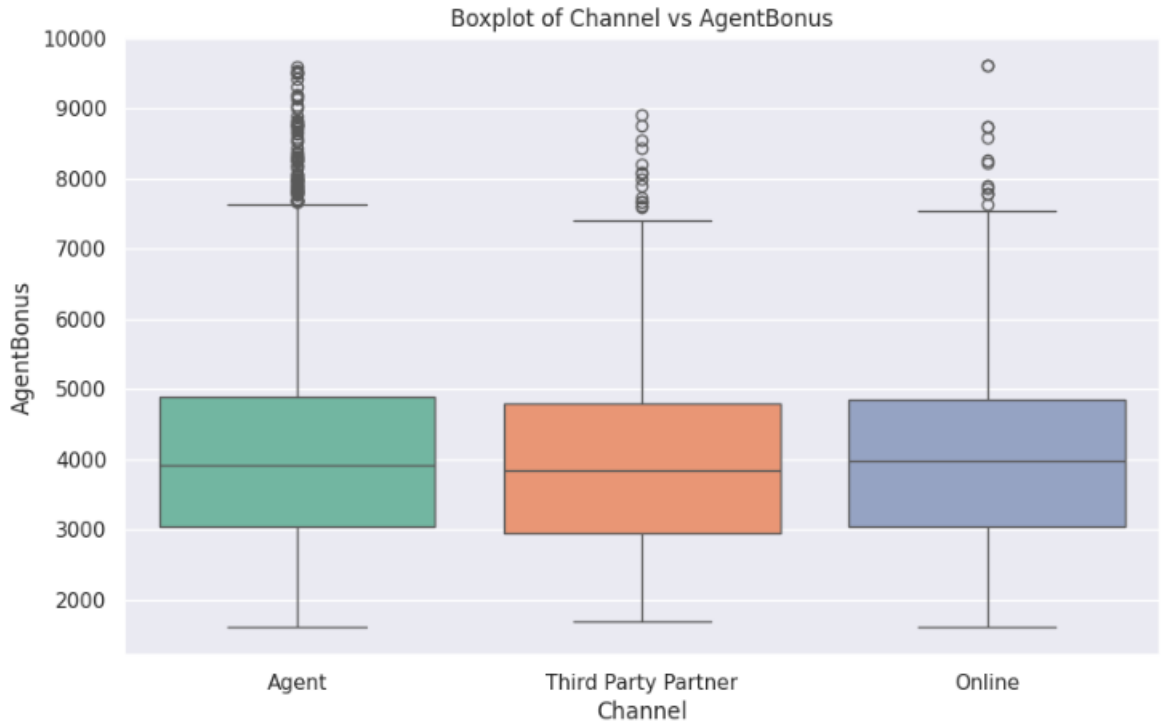

Table 24: Scatterplot of Channel Vs AgentBonus

- The agent bonus median is similar across all the channel.
- Online and Agent channels appear to have a marginally higher median compared to Third Party Partner
- The Interquartile Range (IQR) presented by the boxes is consitant across all three groups, This shows the agents seems to get the bonuses between 3,000 to 5,000.
- Agent Channel has most dense outliers reaching upto 1,00,000.

b) Occupation vs. Agent Bonus

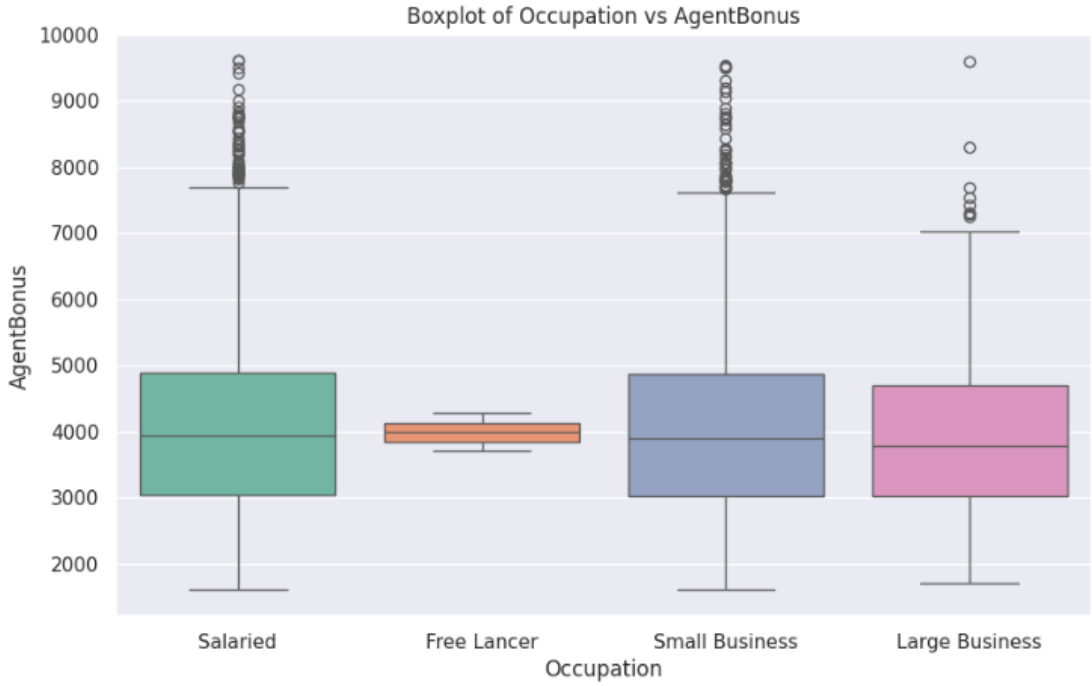

Table 25:Boxplot of Occupation Vs Agent Bonus

- All the four groups has the similar median bonuses which is 4,000.
- The Salaried and small business groups has the highest IQR and the longest whickers while freelancer has the lowest.
- Large business shows lower whisker as well as IQR as compare to salaried and small business.
- Salaried and small business groups has outliers reaching to 1,00,000. Both groups have individuals earning bonuses up to 9,000 - 10,000.
- Large business as well have several high outliers.

c) Designation Vs AgentBonus

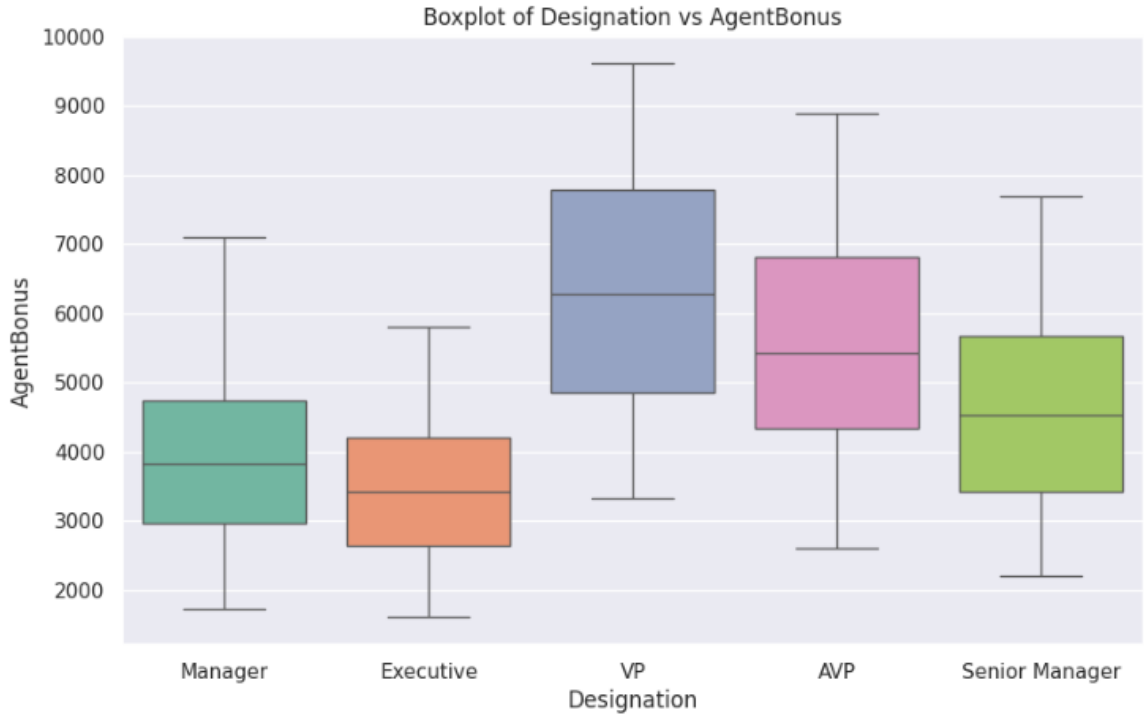

Table 26:Boxplot of Designation Vs AgentBonus

- VP and AVP has the highest median, followed by senior manager.
- The customers with the designation manager and executive shows lower median bonuses of around 3,500 and 4,000.

We have completed the EDA, Univariate, Bivariate and Multivariate analysis. Now let's proceed further with preprocessing the data and model building.

Data Preprocessing

Duplicate value check and treatment

- There are no duplicates present in the data.

Anomalous value check and treatment

During anomalous value analysis, lower age values were observed. However, these values accounted for a substantial portion of the dataset (~3000 records) and are logically valid in the insurance context, where policies may be issued to minors. Therefore, these observations were retained to avoid loss of meaningful information.

Missing value check and treatment

Before:

	0
AgentBonus	0
Age	269
CustTenure	226
Channel	0
Occupation	0
EducationField	0
Gender	0
ExistingProdType	0
Designation	0
NumberOfPolicy	45
MaritalStatus	0
MonthlyIncome	236
Complaint	0
ExistingPolicyTenure	184
SumAssured	154
Zone	0
PaymentMethod	0
LastMonthCalls	0
CustCareScore	52

Table 28: Missing Values Before Treatment

After:

	0
AgentBonus	0
Age	0
CustTenure	0
Channel	0
Occupation	0
EducationField	0
Gender	0
ExistingProdType	0
Designation	0
NumberOfPolicy	0
MaritalStatus	0
MonthlyIncome	0
Complaint	0
ExistingPolicyTenure	0
SumAssured	0
Zone	0
PaymentMethod	0
LastMonthCalls	0
CustCareScore	0

Table 27: Missing Values After Treatment

- Missing values are treated using median for numrical variables.

Outlier check and treatment

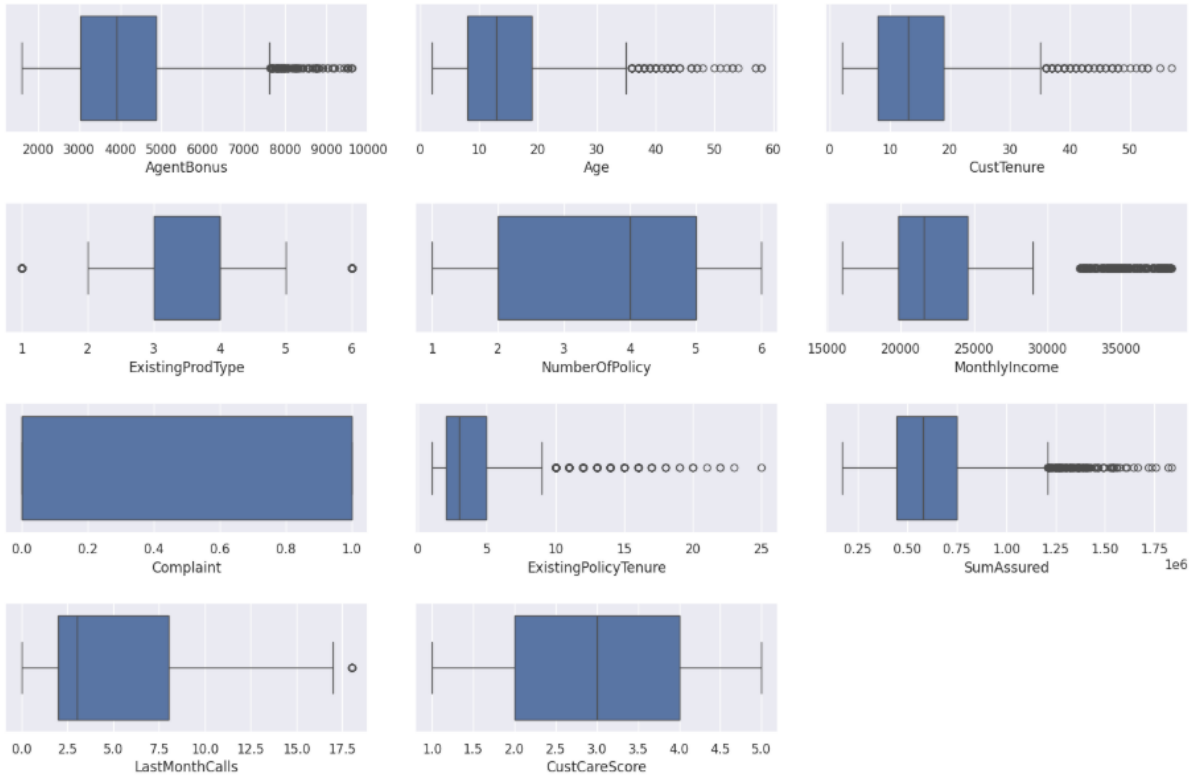

Table 29: Boxplots of all the Numeric Variables

Observation:

- Agent Bonus shows some high-value outliers, which likely represent top-performing agents and they can be considered as valid and important for the model building.
- Age contains higher-end outliers; however, these fall within a realistic range and are retained.
- Customer Tenure shows a few higher values, which are reasonable for long-term customers.
- Existing Product Type shows minimal variation with no significant outlier concern.
- Monthly Income shows higher-end outliers, representing high-income individuals, which is realistic.
- Existing Policy Tenure shows some higher-value outliers, likely representing long-standing policies.
- Sum Assured contains extreme high values, which are valid as they are representing high-value insurance policies.

Outliers were identified using boxplots; however, they were retained as they represent meaningful business variations and not data errors.

Feature Engineering

In the dataset all the variables are meaningful and sufficient for analysis We do not require further transformation, and retaining the original features helped preserve data integrity.

Data preparation for modeling

```
    Age  CustTenure              Channel      Occupation EducationField  \
0  22.0         4.0                Agent        Salaried       Graduate
1  11.0         2.0  Third Party Partner        Salaried       Graduate
2  26.0         4.0                Agent     Free Lancer  Post Graduate
3  11.0        13.0  Third Party Partner        Salaried       Graduate
4   6.0        13.0                Agent  Small Business             UG

   Gender  ExistingProdType Designation  NumberOfPolicy MaritalStatus  \
0  Female                 3     Manager             2.0        Single
1    Male                 4     Manager             4.0      Divorced
2    Male                 4   Executive             3.0     Unmarried
3  Female                 3   Executive             3.0      Divorced
4    Male                 3   Executive             4.0      Divorced

   MonthlyIncome  Complaint  ExistingPolicyTenure  SumAssured   Zone  \
0        20993.0          1                   2.0    806761.0  North
1        20130.0          0                   3.0    294502.0  North
2        17090.0          1                   2.0    578976.5  North
3        17909.0          1                   2.0    268635.0   West
4        18468.0          0                   4.0    366405.0   West

  PaymentMethod  LastMonthCalls  CustCareScore
0   Half Yearly               5            2.0
1        Yearly               7            3.0
2        Yearly               0            3.0
3   Half Yearly               0            5.0
4   Half Yearly               2            5.0
0    4409
1    2214
2    4273
3    1791
4    2955
Name: AgentBonus, dtype: int64
```

Table 30: X and Y Variables

- Defined the X and Y variables.
- Created dummy variables and converted them into float.
- Split the data between 70:30

Model Building - Baseline Model

```
                            OLS Regression Results
==============================================================================
Dep. Variable:             AgentBonus   R-squared:                       0.811
Model:                            OLS   Adj. R-squared:                  0.809
Method:                 Least Squares   F-statistic:                     384.2
Date:                Sat, 28 Mar 2026   Prob (F-statistic):               0.00
Time:                        11:25:58   Log-Likelihood:                -24785.
No. Observations:                3164   AIC:                         4.964e+04
Df Residuals:                    3128   BIC:                         4.986e+04
Df Model:                          35
Covariance Type:            nonrobust
==========================================================================================
                                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                       1143.7466    479.057      2.387      0.017     204.448    2083.045
Age                           22.0890      1.438     15.361      0.000      19.269      24.909
CustTenure                    23.6222      1.449     16.303      0.000      20.781      26.463
ExistingProdType              49.8593     22.516      2.214      0.027       5.712      94.006
NumberOfPolicy                 6.9711      7.941      0.878      0.380      -8.599      22.541
MonthlyIncome                  0.0284      0.005      5.602      0.000       0.018       0.038
Complaint                     48.5814     24.381      1.993      0.046       0.777      96.386
ExistingPolicyTenure          36.2895      3.519     10.313      0.000      29.390      43.189
SumAssured                     0.0034   5.93e-05     57.708      0.000       0.003       0.004
LastMonthCalls                -3.6475      3.266     -1.117      0.264     -10.051       2.756
CustCareScore                 10.4650      8.040      1.302      0.193      -5.299      26.229
Channel_Online                29.7068     36.304      0.818      0.413     -41.475     100.888
Channel_Third Party Partner   11.1390     28.502      0.391      0.696     -44.745      67.023
Occupation_Large Business   -634.5366    464.235     -1.367      0.172   -1544.773     275.700
Occupation_Salaried         -474.3487    438.917     -1.081      0.280   -1334.942     386.245
Occupation_Small Business   -605.4016    447.301     -1.353      0.176   -1482.435     271.632
EducationField_Engineer       18.3957    158.798      0.116      0.908    -292.964     329.755
EducationField_Graduate     -119.5960     92.492     -1.293      0.196    -300.946      61.754
EducationField_MBA          -164.3689    131.380     -1.251      0.211    -421.968      93.230
EducationField_Post Graduate -106.3769   104.766     -1.015      0.310    -311.795      99.041
EducationField_UG             26.2630     62.015      0.423      0.672     -95.331     147.857
EducationField_Under Graduate  -8.5159     39.265     -0.217      0.828     -85.504      68.473
Gender_Male                   29.1273     22.490      1.295      0.195     -14.969      73.224
Designation_Executive       -426.0630     64.380     -6.618      0.000    -552.294    -299.832
Designation_Manager         -396.8905     55.008     -7.215      0.000    -504.746    -289.035
Designation_Senior Manager  -188.7635     52.178     -3.618      0.000    -291.070     -86.457
Designation_VP                16.8708     72.806      0.232      0.817    -125.881     159.623
MaritalStatus_Married        -42.6401     30.218     -1.411      0.158    -101.888      16.608
MaritalStatus_Single          40.2649     33.389      1.206      0.228     -25.202     105.731
MaritalStatus_Unmarried     -200.7329     62.587     -3.207      0.001    -323.450     -78.016
Zone_North                    55.7339     94.919      0.587      0.557    -130.376     241.844
Zone_South                   205.4121    291.162      0.705      0.481    -365.476     776.300
Zone_West                     60.7281     94.391      0.643      0.520    -124.347     245.803
PaymentMethod_Monthly        170.6243     60.433      2.823      0.005      52.132     289.117
PaymentMethod_Quarterly      102.7014     89.166      1.152      0.249     -72.128     277.531
PaymentMethod_Yearly         -88.9275     34.805     -2.555      0.011    -157.170     -20.685
```

Table 31: OLS Baseline Regression Model

Regression Results:

- R-squared is 0.81 which means the model explains 81% of the variation in Agent Bonus, indicating a strong predictive ability.
- Adjusted R-squared is 0.80 closed to R-squared, suggesting that most variables included in the model are relevant and not adding unnecessary complexity.
- Key factors such as Age, Customer Tenure, Number of Policies, Monthly Income, and Existing Policy Tenure significantly influence Agent Bonus.

The overall model seems to be good.

Performance of the model on multiple metrics

Linear Regression Model Training performance

Training Performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	610.65828	479.567344	0.811281	0.809108	12.476191

Table 32: Linear Regression Training Performance

Linear Regression Model Training performance

Testing Performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	625.773096	494.254348	0.799406	0.793931	12.858953

Table 33: Linear Regression Testing Performance

Performance Observation:

- The model explains around 80-81% of the variation in agent bonus.
- The RMSE value (625) suggests that the average prediction error is moderate and acceptable in the context of bonus prediction.
- The MAE (494) indicates that, on average, predictions deviate by around 494 units from actual values.
- The train and test RMSE and MAE are low and comparable so model is not overfitting.
- The MAPE on the test set suggests that model's predictions are off by about 12.8% from the actual values which is a good.

Model Building - Advanced Models

After establishing the baseline model, now we can develop advanced machine learning models to improve prediction accuracy and capture complex relationships within the data. These models can be evaluated on both the training and test datasets using multiple performance metrics to ensure robustness and generalization.

Decision Tree Model Regressor:

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.0	0.0	1.0	1.0	0.0

Table 34: Decision Tree Model performance on train set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	747.616992	537.5059	0.713686	0.705871	13.887658

Table 35: Decision Tree Model Performance on test set

- The Decision Tree regressor is highly overfitting the training data but much lower performance on test data.
- As per the performance on test set model explains approximately 71.4% of the variance, indicating a moderate predictive performance.
- RMSE (747) and MAE (537) are high and MAPE (13.9%) indicates that predictions are off by about 14% on average.

Bagging Regressor:

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	234.318638	164.036757	0.972213	0.971894	4.222224

Table 36: Bagging Model performance on Training set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	569.614837	426.380973	0.833794	0.829257	11.127216

Table 37:Bagging model performance on test set

- Model is moderately overfitting as training performance is much better than test performance
- Model explains 97% of the variance also we have low MAPE (4.2%) on train set.
- R2 dropped from 97% to 83% on test set also MAPE increased to 11%.

Random Forest Model Regressor:

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	193.793838	145.265164	0.980994	0.980775	3.79022

Table 38:Random Forest Model train performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	532.925502	404.293363	0.854515	0.850544	10.556132

Table 39: random Forest Model Test performance

- Model is overfitting as performed good on train and weak on test set.
- Very high R-squared(98%) means model explains almost all variance on the train set also the MAPE is lower which is 3.7%
- R-squared dropped from 98% to 85% on the test set and MAPE has increased from 3.7% to 10.56%

AdaBoost Regressor:

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	689.330073	585.379451	0.759522	0.756754	16.307838

Table 40: AdaBoost Model Performence on Train set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	713.330367	609.427328	0.739345	0.732231	16.880758

Table 41: AdaBoost Model performance on Test set

- Model is slightly underfitting as the R-squered difference between train and test set is very small
- Model expains (R-squered = 75%) of the variance on train set also the same has decreased to (R-squered = 73%) on test set. MAPE for both are relatively high.
- The RMSE (713) and MAE (~609) are also high on test set, showing that the model makes large prediction errors.

Gradient Boosting Regressor

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	490.923403	382.948882	0.878031	0.876627	10.042787

Table 42: Gradiant Boosting Model performance on train set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	547.099154	425.80001	0.846674	0.842489	11.058996

Table 43: Gradiant Boosting model performance on test set

- model shows good performance, with training R2 of 88% and test R2 of 85%, indicating strong predictive capability.
- The RMSE (491 train, 547 test) and MAE (383 train, 426 test) are moderate and relatively close.
- The small difference between training and test results indicates minimal overfitting and good generalization.
- The MAPE shows 10% and 11% for both that predictions are off by around 10%

XGBoost Regressor:

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	155.736268	113.248802	0.987726	0.987584	3.102253

Table 44: XGBoost Model Performance on train set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	577.100403	441.875641	0.829397	0.82474	11.510718

Table 45: XGBoost Model Performance on test set

- The model shows excellent performance on training data with R2 of 98% with very low MAPE (3%)
- However on the test set performance drops to 82% with high MAPE (11.51%)
- This large gap between training and test performance indicates overfitting.

After building advanced models, **Decision Tree, Random Forest, and Gradient Boosting** were selected for further analysis through Hyperparameter tuning. These models exhibited signs of overfitting with default parameters, indicating high variance. Therefore, Hyperparameter tuning was performed to control overfitting and improve their generalization performance.

Model Performance Improvement using Hyperparameter Tuning

As our previous Advanced models were overfitting and where performance on training data was significantly better than on test data. Hyperparameter tuning helps in controlling model complexity, thereby reducing overfitting and improving generalization.

Hyperparameter Tuning - Decision Tree:

Grid Search Cross-Validation was used to optimize model parameters, improving prediction accuracy and ensuring robust performance.

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	501.580473	370.445669	0.872679	0.871213	9.875297

Table 46: Decision Tree Tuned Model performance on Train set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	602.477425	458.53821	0.814063	0.808988	11.873252

Table 47:Decision Tree Tuned Model performance on Test set

- Model is generalizing better than the decision tree model with default parameters
- The overfitting has reduced so much after hyperparameter tuning.
- Model explains about 87.3% of variance on train set also MAPE is lower than 10% which highly accurate for prediction.
- Model explains about 81.4% of variance on test data.

Hyperparameter Tuning - Random Forest:

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	192.725098	144.819769	0.981203	0.980986	3.779326

Table 48: Random Forest Model Tuned performance on Train set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	531.37077	403.091556	0.855363	0.851415	10.515849

Table 49: Random Forest Model tuned model performance on test set

- No significant change in the tuned model as compared to the model with default parameters.
- Model explains 98% variance on train data the same is decreased to R2 = 85% in test set.

Hyperparameter Tuning - Gradient Boosting Regressor:

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	332.93981	254.017439	0.943901	0.943256	6.921347

Table 50: Gradient Boosting Tuned Model performance on Train set

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	538.198813	406.817592	0.851622	0.847572	10.567259

Table 51:Gradient Boosting Tuned Model performance on Test Set

- The R-Squared has increased from 87% to 94%, Model explains variance of 94%.
- MAPE has also decreased from 10% to 6.9%
- Also the test performance has been improved from 84% to 85% after tuning.
- The scores of the test and train data has been improved after Hyper parameter tuning.
- The model has reduced overfitting and shows good generalization.

Hyperparameter tuning reduced overfitting and improved model generalization, resulting in more accurate and reliable predictions. So we can now proceed with comparison of all the model performance and Final Model Selection.

Model Performance Comparison and Final Model Selection

	Decision Tree	Decision Tree Tuned	Random Forest Estimator	Random Forest Tuned	Gradient Boost Estimator	Gradient Boost Tuned
RMSE	0.0	501.580473	193.793838	192.725098	490.923403	332.939810
MAE	0.0	370.445669	145.265164	144.819769	382.948882	254.017439
R-squared	1.0	0.872679	0.980994	0.981203	0.878031	0.943901
Adj. R-squared	1.0	0.871213	0.980775	0.980986	0.876627	0.943256
MAPE	0.0	9.875297	3.790220	3.779326	10.042787	6.921347

Table 52: Training Performance Comparison

	Decision Tree	Decision Tree Tuned	Random Forest Estimator	Random Forest Tuned	Gradient Boost Estimator	Gradient Boost Tuned
RMSE	747.616992	602.477425	532.925502	531.370770	547.099154	538.198813
MAE	537.505900	458.538210	404.293363	403.091556	425.800010	406.817592
R-squared	0.713686	0.814063	0.854515	0.855363	0.846674	0.851622
Adj. R-squared	0.705871	0.808988	0.850544	0.851415	0.842489	0.847572
MAPE	13.887658	11.873252	10.556132	10.515849	11.058996	10.567259

Table 53: Testing Performance Comparison

- The Decision Tree model shows severe overfitting with perfect training performance (R2 = 1.0) but significantly lower test performance (R2 = 0.71) after hyperparameter tuning improved model generalization, especially for Decision Tree and Gradient Boosting model
- Random Forest Tuned achieved the best performance with highest test R2 = 85% and lowest RMSE (531), indicating strong predictive accuracy.

We came to conclusion that amongst all the estimator and **Random Forest Tuned Model and Gradient Boosting Tuned** Model performed similarly with the highest test R-squared (85%) and lower error values, making them the best-performing models, while Decision Tree showed comparatively weaker performance.

Both are very close but random forest tuned model shows more stable and overall better performance GB very closed but marginally behind.

So we are choosing **Random Forest Tuned Model** as the best model.

Checking important features of the Random Forest Tuned Model

Feature importance helps to analyze and visualize the key drivers of the Bonus prediction also it will help us to know the which variables have the highest impact agent performance and rewards.

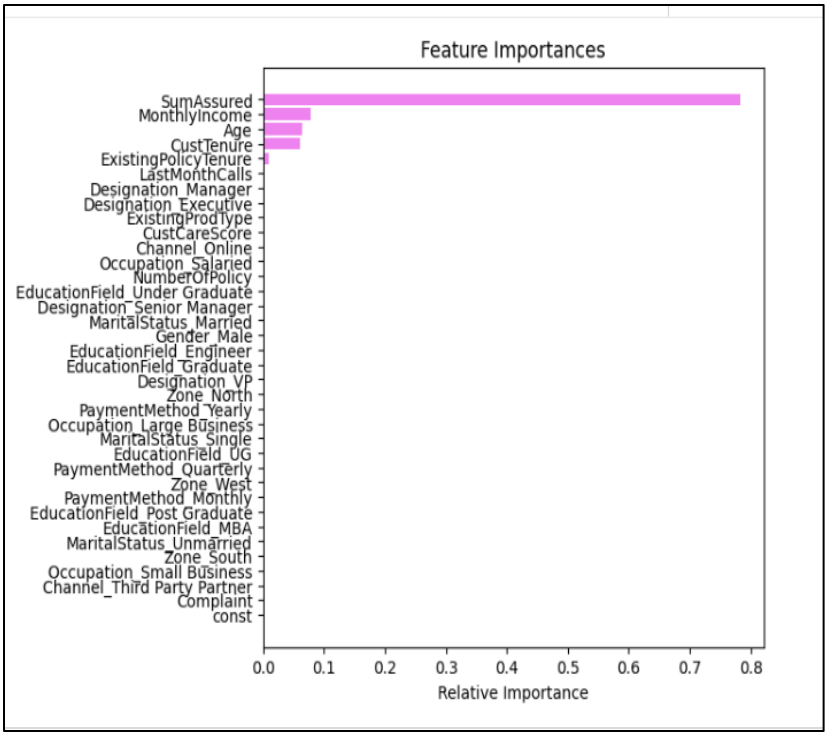

Table 54: Feature Importance

- As per the analysis the most important features in Random Forest Tuned model are Sum Assured, Monthly Income, Age, Customer Tenure and Existing Policy Tenure.
- SumAssured is the most influential feature, indicating that higher policy values drive agent bonuses, while other factors have comparatively lower impact.

Observed VS Predicted Values Plot of the test data for the Random Forest Model (Tuned)

Also the below plot can help us to visualize how closely predicted values match actual values, indicating the model's accuracy and any prediction errors.

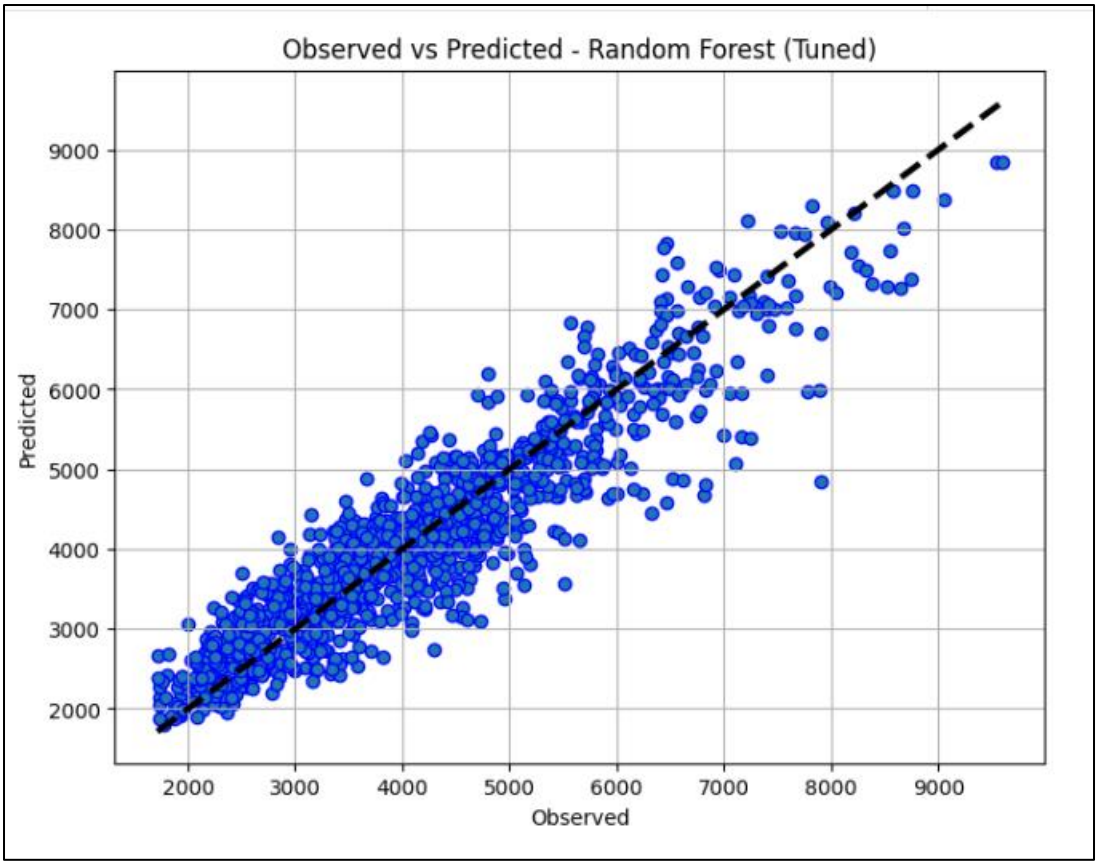

Table 55:Observed Vs Predicted Values Plot

- The plot indicates that predicted values closely follow the actual values, showing good model accuracy with minor deviations at higher ranges.

Key insights from the analysis conducted

- After the analysis the final model (Random Forest Tuned) is able to explain 85% of the variation in agent bonuses (R2 = 0.85), indicating a strong ability to predict bonus outcomes and support decision-making.
- Among all models, Random Forest provides the most accurate and stable predictions with the lowest error (RMSE = 531, MAE = 403), making it suitable for consistent and scalable business use.
- The relatively small gap between training and test performance in ensemble models indicates that the model is not heavily overfitted and can generalize well to new data, ensuring reliability in real-world scenarios.
- MAPE were reduced after tuning across all models, with Random Forest achieving the lowest MAPE of approximately 10.5%, indicating predictions are, on average, within 10% of actual values.

Actionable insights and business recommendations

- The company can use this forecasting model to predict Agent Bonuses which will be helping in efficient incentive allocation amongst Agent based on their overall performance.
- As Sum Assured contributing 78% of the bonus prediction company should foucus on redesigning structure of the rewards for selling higher value policies, Introducing tier-based bonuses (i.e higher incentives for policies above certain thresholds) can help in business growth as well.
- As the second most import features is Monthly income indicate that higher income customers are more valuable, The company should prioritize these segments through targeted marketing campaigns and personalized offerings,
- Using the model's predictions, A company can categorized agents into high, medium, and low performers.

  High performers - reward with premium incentives and retention programs,

  Low performers - assign structured training and mentoring

  This ensures efficient allocation of training and incentive resources.
- Also through the analysis its been observed that different age group varying insurance needs—for example, younger customers may prefer long-term investment-linked policies, while older customers may prioritize security focused plans. Aproaching insurance plans accordingly can help agents target sales and maximise bonus outcomes.
- Customers with longer relationships are more likely to trust the company and purchase additional policies. Agents can focus on existing customers for cross-selling and upselling opportunities.

Conclusion

The project successfully developed predictive model which is Tuned Random Forest model with high accuracy and strong generalization on test data. Hyperparameter tuning played a key role in reducing overfitting and improving overall model performance.

The tuned Random Forest model achieved the best performance (R2 - 85%) and was selected as the final model. The analysis revealed that Sum Assured is the key driver of agent bonuses, supported by factors like income, age, and tenure. The model enables data-driven bonus prediction, improved sales strategies, and efficient resource allocation, ultimately enhancing overall business performance.